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FINAL PROJECT REPORT (FPR)

Comparative Literature Review and Interactive Dashboard on Global AI Healthcare Regulations

Comparative Compliance Analytics across 20 Jurisdictions

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Résumé

L'intelligence artificielle (IA) transforme rapidement le secteur de la santé à travers l'imagerie diagnostique, la pathologie numérique, la génomique, l'aide à la décision clinique et la découverte de médicaments. Cette transformation s'accompagne d'un paysage réglementaire fragmenté: les exigences en matière de logiciel dispositif médical (SaMD), de protection des données de santé, de validation clinique, de transparence algorithmique et de surveillance post-marché varient fortement selon les juridictions. Pour les développeurs, les hôpitaux et les décideurs publics, cette complexité multiplie les coûts d'accès au marché et les risques de non-conformité.

Ce Projet de Fin d'Études pose la question suivante: comment synthétiser, noter et comparer de manière systématique les exigences réglementaires de l'IA en santé à travers plusieurs juridictions? La démarche combine (1) une revue de littérature thématique couvrant les cadres réglementaires et éthiques; (2) un jeu de données curé de vingt pays/régions notés sur sept dimensions de conformité; (3) un tableau de bord interactif (Streamlit), publié en open source, permettant la comparaison géospatiale, thématique et par cas d'usage clinique. Les résultats mettent en évidence une convergence vers la classification par risque et la protection des données inspirée du RGPD, tout en soulignant des écarts persistants de capacité d'application entre économies avancées et émergentes. L'EU AI Act apparaît comme un point d'inflexion pour la gouvernance contraignante de l'IA à haut risque en santé. L'outil est conçu comme support de décision et de recherche—et non comme conseil juridique—avec une priorité à la reproductibilité.

Mots-clés: régulation de l'IA en santé, SaMD, conformité, RGPD, EU AI Act, validation clinique, tableau de bord, revue systématique.

Abstract

Artificial intelligence (AI) is transforming healthcare through diagnostic imaging, digital pathology, genomics, clinical decision support, and drug-discovery workflows. The same properties that make clinical AI powerful—adaptivity, data dependence, and sometimes opacity—also create novel safety, privacy, fairness, and accountability risks. Across jurisdictions, Software as a Medical Device (SaMD) pathways, health-data protection rules, clinical validation expectations, algorithmic transparency obligations, and post-market surveillance duties differ substantially. The resulting regulatory fragmentation raises market-entry costs for developers, procurement uncertainty for hospitals, and comparative-learning barriers for policymakers.

This Final Year Project addresses the research question: how can multi-jurisdictional AI healthcare regulatory requirements be systematically synthesized, scored, and interactively compared? The work delivers three integrated contributions. First, a thematic literature and policy synthesis organizes prior work around seven compliance dimensions rather than paper-by-paper summary. Second, a curated dataset covers twenty countries and regions, each scored on an ordinal 1–10 scale for data privacy, clinical validation, approval process, transparency, ethics, post-market surveillance, and liability. Third, an interactive Streamlit dashboard—published openly at GitHub—supports geospatial maps, radar comparisons, theme heatmaps, trend cards, country narratives, and literature browsing from the same JSON source of truth.

Findings indicate global convergence toward risk-based device classification and GDPR-inspired data protection, alongside persistent enforcement-capacity gaps between advanced and emerging economies. Liability and transparency frequently lag privacy and market-access themes. The EU AI Act emerges as a watershed for binding high-risk healthcare AI governance, while the United States illustrates high authorization throughput under a more sectoral regime. The dashboard is a decision-support and research instrument, not legal advice, and is designed for reproducibility through version-controlled data and open source code.

Keywords: AI healthcare regulation, SaMD, compliance dashboard, GDPR, EU AI Act, clinical validation, post-market surveillance, comparative governance.

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List of Abbreviations and Acronyms

Abbreviation	Meaning
AI / ML / DL	Artificial Intelligence / Machine Learning / Deep Learning
API	Application Programming Interface
CDS	Clinical Decision Support
DPIA	Data Protection Impact Assessment
EHDS	European Health Data Space
EMA	European Medicines Agency
FDA	U.S. Food and Drug Administration
GDPR	General Data Protection Regulation
GMLP	Good Machine Learning Practice
HIPAA	Health Insurance Portability and Accountability Act
HITL	Human-in-the-Loop
IMDRF	International Medical Device Regulators Forum
KPI	Key Performance Indicator
LLM	Large Language Model
MDR	Medical Device Regulation (EU)
MHRA	Medicines and Healthcare products Regulatory Agency (UK)
NLP	Natural Language Processing
NMPA	National Medical Products Administration (China)
PCCP	Predetermined Change Control Plan
PFE	Projet de Fin d'Études (Final Year Project)
PGE	Grande École Programme
PMDA	Pharmaceuticals and Medical Devices Agency (Japan)
RWE / RWD	Real-World Evidence / Real-World Data
SaMD	Software as a Medical Device
SLR	Systematic Literature Review
TPLC	Total Product Lifecycle
WHO	World Health Organization
XAI	Explainable Artificial Intelligence

1. Introduction

1.1 General Context and Societal Stakes

Healthcare systems worldwide face simultaneous pressures of ageing populations, clinician shortages, rising chronic disease burden, and escalating cost of care. Artificial intelligence (AI) has emerged as a candidate technology to improve diagnostic accuracy, accelerate triage, personalize therapy, and automate administrative workflows [1], [2]. In imaging alone, AI-enabled Software as a Medical Device (SaMD) products have proliferated: the U.S. Food and Drug Administration (FDA) has authorized hundreds of AI/ML-enabled devices, signaling industrial maturity and clinical appetite [3], [4].

Yet the same properties that make AI powerful—adaptivity, opacity, and dependence on large training corpora—also create novel safety, privacy, fairness, and accountability risks [5], [6]. Unlike a traditional catheter or implant, an adaptive clinical model can drift after deployment as local patient distributions shift. Unlike a deterministic calculator, a deep neural network may resist clinician-facing explanation [7], [8]. Unlike many consumer applications, healthcare AI frequently processes special-category personal data under stringent legal regimes such as the EU General Data Protection Regulation (GDPR) and the U.S. Health Insurance Portability and Accountability Act (HIPAA) [9], [10].

The result is what this project terms *regulatory lag*: technology cycles outpace legislative cycles, producing a multi-jurisdictional "maze" of overlapping medical-device, data-protection, AI-specific, and ethical frameworks. For developers seeking multi-market access, for hospitals evaluating procurement risk, and for policymakers seeking evidence-based comparative learning, the absence of a structured, explorable synthesis is itself a barrier to safe innovation.

1.2 Micro Context and Project Motivation

The present work addresses comparative AI healthcare compliance through two complementary deliverables:

1. a thematic, multi-region literature and policy synthesis organized around seven compliance dimensions;
2. an interactive analytics dashboard that transforms curated jurisdictional indicators into maps, comparisons, trend views, and clinical use-case readiness profiles, released openly at `https://github.com/remi7025/AI-Healthcare-Compliance-Dashboard`.

The Phase 1 proposal framed the problem as regulatory fragmentation, safety/ethics obligations for dynamic systems, and market-entry barriers created by tightening 2025–2026 documentation and governance standards. It further argued that Natural Language Processing (NLP) and human-in-the-loop (HITL) workflows can reduce rote scanning of legislative corpora. The implemented draft focuses first on high-quality curated comparative analytics and interactive visualization, while specifying a reproducible NLP-assisted ingestion pipeline as the methodological backbone for future automation.

1.3 Problem Statement

The central research and operational question of this PFE is:

> *How can multi-jurisdictional AI healthcare regulatory requirements be systematically synthesized, scored, and interactively compared so that researchers and decision-makers can identify maturity gaps, convergence trends, and deployment constraints across countries?*

This question decomposes into four concrete challenges:

- **Knowledge fragmentation:** peer-reviewed studies, agency guidance, and national AI strategies are dispersed across databases and languages;
- **Incommensurable frameworks:** FDA Class I–III, EU MDR classes, and NMPA tiers are related but not identical;
- **Static reporting:** traditional literature reviews freeze knowledge at publication time and do not support drill-down exploration;
- **Stakeholder diversity:** policymakers, developers, clinicians, and researchers need different views of the same underlying evidence.

1.4 SMART Objectives and Contributions

The project pursues the following SMART objectives:

1. **Specific:** synthesize AI healthcare regulation across twenty countries/regions spanning six geographic areas;
2. **Measurable:** define and score seven compliance themes on a calibrated 1–10 scale and report device-approval counts where available;
3. **Achievable:** deliver a version-controlled JSON dataset and interactive dashboard runnable locally without proprietary APIs;
4. **Relevant:** address an industry-critical need for comparative compliance intelligence under EU AI Act and related regimes;
5. **Time-bound:** produce Phase 1 framing, literature/dataset/dashboard drafts, and this full report on the planned project schedule.

Scientific and technical contributions claimed by this work:

- a thematic state-of-the-art that organizes regulatory literature by problem strand rather than paper-by-paper summary;
- an explicit seven-theme scoring ontology linking privacy, validation, approval, transparency, ethics, post-market, and liability;

- a reproducible dashboard architecture (Streamlit prototype and React presentation layer) with exportable tables;
- critical positioning against gaps in gold-standard cross-jurisdictional compliance repositories.

1.5 Scope, Assumptions, and Non-Goals

The analysis covers twenty jurisdictions selected for regional diversity and data availability (United States, Canada, European Union aggregate, United Kingdom, Germany, Switzerland, China, India, Japan, South Korea, Singapore, Thailand, Saudi Arabia, UAE, Israel, South Africa, Nigeria, Kenya, Australia, and Brazil). Theme scores are expert-curated comparative indicators informed by policy documents and literature; they are *not* official government ratings and must not be interpreted as legal advice.

Out of scope for the present draft: live scraping of every national gazette; clinical validation of a therapeutic AI model; production deployment inside a hospital EHR; and binding legal opinions. NLP classification is designed and partially prototyped as a pipeline, with full supervised training treated as future work when labeled corpora mature.

1.6 Report Outline

Chapter 2 presents the state of the art on SaMD frameworks, privacy, validation, transparency, ethics, surveillance, and liability, closing with comparative positioning. Chapter 3 situates the mission, role, data accessed, and legal constraints. Chapter 4 details the reproducible methodology: search strategy, scoring protocol, dataset schema, NLP pipeline design, and dashboard architecture. Chapter 5 reports quantitative and qualitative findings, regional patterns, and critical discussion. Chapter 6 answers the research question, states limitations, and proposes future work. Appendices provide code extracts, environment details, and a declaration of generative-AI use.

1.7 Stakeholder Needs Analysis

Four primary stakeholder groups motivate the dashboard design. **Researchers** need citable thematic synthesis and exportable tables. **Policymakers** need rapid gap identification across peer jurisdictions. **Medtech/AI developers** need early visibility into documentation and evidence expectations before market sequencing. **Hospital compliance and clinical-informatics teams** need an interactive artifact that makes comparative governance tangible. Each group implies different default views: literature-first, heatmap-first, radar comparison-first, and guided tour-first, respectively. The implemented interface exposes all four without forcing a single narrative path.

1.8 Project Success Criteria

Project-level success is defined as: (i) complete theme vectors for all selected jurisdictions; (ii) a runnable local demo from a clean environment; (iii) a bibliography grounded in peer-reviewed and primary regulatory sources; (iv) an explicit positioning statement against related work; and (v) clear legal caveats that comparative scores are research instruments, not legal advice.

1.9 Key Takeaways

Readers short on time may retain three messages: (1) healthcare AI regulation is converging on risk-based SaMD thinking and GDPR-like privacy, yet diverging on transparency mandates and liability; (2) open comparative scorecards plus interactive analytics make those patterns inspectable; (3) automation should accelerate document triage under human accountability, not replace it. The remainder of the report supplies the evidence, methods, and limitations behind those messages.

2. State of the Art

This chapter organizes prior work by thematic strands rather than paper-by-paper summary. It prioritizes recent sources (approximately 2018–2026) while retaining seminal frameworks such as IMDRF SaMD guidance [11].

2.1 Theoretical Foundations of AI in Healthcare

2.1.1 From Statistical Learning to Clinical AI Systems

Modern clinical AI draws on supervised learning, deep neural networks, and increasingly foundation models. In imaging, convolutional architectures and transformers support detection, triage, and quantification tasks [2]. In text-heavy domains—radiology reports, guidelines, and regulatory corpora—transformer language models enable semantic retrieval and classification. The clinical setting, however, imposes constraints rarely present in general ML benchmarks: safety-critical false negatives, shifting populations, missing labels, and medico-legal accountability [1], [6].

2.1.2 Software as a Medical Device (SaMD)

The International Medical Device Regulators Forum (IMDRF) defined SaMD as software intended for medical purposes that is not part of a hardware medical device [11]. Risk categorization considers (i) the significance of the information provided to the healthcare decision and (ii) the state of the healthcare situation or condition. This conceptual scaffolding underpins FDA pathways (510(k), De Novo, PMA), EU MDR classification, Japan's DASH framework, and China's NMPA guiding principles [4], [12], [13], [14].

Table 2.1: Representative risk-based classification systems for medical devices / SaMD

Jurisdiction	System	Classes
United States	FDA risk-based	I, II, III
European Union	MDR risk-based	I, IIa, IIb, III
China	NMPA three-tier	I, II, III
Canada	Health Canada	I, II, III, IV
Japan	PMDA / PMD Act	I, II, III, IV
Australia	TGA	I, IIa, IIb, III

Table 2.1 shows near-universal adoption of proportional regulation by potential harm. Higher-risk autonomous diagnosis and treatment-planning systems face stricter clinical evidence and post-market obligations [15].

2.1.3 Adaptive Algorithms and Lifecycle Thinking

Unlike static devices, AI/ML systems may learn after deployment. Regulators therefore shift from point-in-time approval toward lifecycle governance [16], [17]. The FDA Predetermined Change Control Plan (PCCP) allows manufacturers to pre-specify anticipated modifications and associated validation methods [18]. The EU AI Act imposes ongoing conformity and risk-management duties for high-risk systems [14]. Joint Good Machine Learning Practice (GMLP) principles from FDA, Health Canada, and MHRA articulate ten development practices covering data quality, reference standards, bias mitigation, and monitoring [19].

2.2 Regulatory Frameworks for AI Medical Devices

2.2.1 United States

The FDA AI/ML SaMD Action Plan (2021) consolidates a tailored regulatory approach: pathway clarity, good machine learning practice, bias reduction, transparency, and real-world performance monitoring [4]. The U.S. ecosystem combines deep agency capacity with fragmented federal AI legislation and sectoral privacy (HIPAA) rather than omnibus data protection [9]. Innovation throughput is high—device authorizations exceed those of most peers—but transparency and ethics remain comparatively less mandated than in the EU.

2.2.2 European Union

Europe layers MDR/IVDR device rules, GDPR data protection, and the EU AI Act's risk-tiered AI obligations [14]. Most healthcare AI falls into the high-risk category, triggering technical documentation, logging, human oversight, and data-governance duties. The proposed European Health Data Space (EHDS) aims to unlock secondary use of health data under controlled conditions. Strengths include rights-based coherence; challenges include notified-body capacity and compliance complexity for SMEs.

2.2.3 United Kingdom (Post-Brexit Divergence)

The UK retains GDPR-derived data protection while pursuing a "pro-innovation" AI framework that relies on existing sector regulators rather than a single horizontal AI Act analogue. MHRA SaMD programmes and NICE evidence standards for digital health shape clinical adoption. Dual-track EU/UK compliance is now a practical reality for vendors.

2.2.4 East Asia

China's NMPA AI medical device guiding principles (2021) and algorithm-related rules create a state-guided, rapidly evolving regime with significant data-localization implications [13]. Japan's PMDA DASH framework and South Korea's high approval throughput illustrate alternative paths combining industrial policy and regulatory science [12]. Cross-border validation remains sensitive where health data export is restricted.

2.2.5 Emerging and Developing Jurisdictions

Singapore's Model AI Governance Framework and AI Verify testing tools position it as a governance laboratory. India's DPDP Act and Ayushman Bharat Digital Mission create foundational infrastructure with still-maturing AI-device pathways. Gulf states invest heavily in AI strategies while regulatory maturity catches up. African jurisdictions such as South Africa (POPIA), Nigeria (NDPA), and Kenya have advanced data protection ahead of AI-specific SaMD guidance, reflecting capacity constraints and competing health priorities.

2.3 Data Privacy and Governance

2.3.1 GDPR as Global Benchmark

GDPR principles—lawfulness, fairness, transparency, purpose limitation, minimization, accuracy, storage limitation, integrity, and accountability—have become the de facto template for health-data governance worldwide. Article 22's automated decision-making provisions intersect directly with clinical AI [10]. Data Protection Impact Assessments (DPIAs) are expected for high-risk processing.

2.3.2 Post-GDPR Legislative Wave

Since 2018, comprehensive laws have proliferated (e.g., Brazil LGPD, India DPDP, China PIPL, South Africa POPIA, Nigeria NDPA, Thailand PDPA, Saudi PDPL, UAE Federal Decree 45, Swiss nFADP), with varying alignment to GDPR. Cross-border transfer rules and localization requirements complicate multinational training and multi-site clinical validation [5].

2.3.3 Health-Specific Regimes

HIPAA governs U.S. Protected Health Information with Privacy, Security, and Breach Notification Rules [9]. National digital health record laws (e.g., Australia My Health Records) add further constraints. For AI developers, privacy is not a single checklist but a portfolio of overlapping obligations keyed to data residency, consent/legal basis, and secondary research use.

2.4 Clinical Validation and Safety

2.4.1 Evidence Hierarchy for SaMD

Analytical validation, clinical validation, and clinical utility form a common evidence ladder. Risk proportionality determines the depth of prospective trials versus retrospective studies and Real-World Evidence (RWE) [3]. Germany's DiGA pathway and NICE digital evidence standards illustrate pragmatic acceptance of planned evidence generation for lower-risk digital tools.

2.4.2 Bias, Fairness, and Subgroup Performance

Empirical studies document disparate performance across skin tones, demographic groups, and underserved populations [20], [21]. Regulators increasingly expect representative datasets, subgroup analyses, and post-deployment monitoring. From a compliance-analytics perspective, "clinical validation" scores should therefore capture not only the existence of guidance but also maturity of bias-assessment expectations.

2.5 Algorithmic Transparency and Explainability

Transparency spans mandated documentation (EU AI Act technical files, logging, user notices) and voluntary frameworks (NIST AI RMF, Singapore guidance) [14], [22]. The XAI literature warns that post-hoc explanations (SHAP, LIME, attention maps) can create false confidence [7]. Rudin argues for inherently interpretable models in high-stakes decisions [8]. Healthcare regulation must balance clinician trust, patient rights, and the empirical reality that many high-performing models remain partially opaque.

2.6 Ethical Frameworks and the Principles-to-Practice Gap

Jobin et al. mapped a global explosion of AI ethics guidelines [23]. WHO's 2021 guidance articulates six principles: autonomy, well-being/safety, transparency, responsibility, inclusiveness, and sustainability [24]. Mittelstadt cautions that principles alone cannot guarantee ethical AI [25]; Morley et al. survey tools attempting to operationalize ethics [26]. Regional philosophical traditions (EU rights-based, U.S. innovation-oriented, Chinese state-guided, African Ubuntu-informed) shape how principles are interpreted [24].

2.7 Post-Market Surveillance and Liability

Adverse-event systems (FDA MedWatch, EU MDR serious incidents, UK Yellow Card) were designed for hardware-centric failure modes. AI-specific harms—silent performance degradation, systematic bias—may be under-captured [27]. Liability remains largely grounded in product liability and malpractice doctrines that struggle with opaque causation [28]. Proposed EU AI liability reforms would shift evidentiary burdens in some cases, illustrating the frontier nature of this theme in our scoring ontology.

2.8 Related Work on Comparative Regulation and Compliance Tools

Academic comparative studies of AI medical-device approvals (e.g., U.S. vs Europe) quantify throughput and pathway differences [15]. Policy observatories (OECD.AI) and ethics surveys provide transversal maps but rarely couple machine-readable scores with interactive drill-down for healthcare-specific themes [23], [29]. Commercial regulatory intelligence platforms exist for pharma and medtech, yet open, machine-readable datasets with healthcare-specific theme ontologies remain scarce—a gap this project targets.

On the NLP side, transformer-based classification and retrieval have been applied to legal and biomedical text. Systematic review automation using PRISMA-inspired workflows and HITL verification is an active research area. Phase 1 of this project explicitly positioned BERT/transformer pipelines for semantic capture of validation and privacy clauses, while acknowledging that a gold-standard cross-jurisdictional compliance corpus is still underdeveloped.

2.9 Comparative Synthesis Table

Table 2.2: Comparative synthesis of major regulatory approaches (simplified)

Axis	**USA**	**EU**	**China**
Device pathway	510(k)/De Novo/PMA + PCCP	MDR + notified bodies	NMPA AI guiding principles
AI-specific law	Sectoral + EO/NIST voluntary	Binding AI Act high-risk duties	Algorithm/filing rules + industrial policy
Privacy	HIPAA (sectoral)	GDPR + EHDS trajectory	PIPL + localization
Transparency	Encouraged, limited mandates	Strong documentation/logging	Filing + user-facing rules (context-dependent)
Ethics posture	Innovation-oriented voluntary	Rights-based enforceable	State-guided harmony/collective framing

Table 2.2 is intentionally schematic: national systems evolve quickly, and "EU" aggregates heterogeneous member-state enforcement. The dashboard operationalizes a finer seven-theme scorecard to avoid single-axis rankings.

2.10 Positioning of This Work

Relative to prior literature, this PFE:

1. integrates *policy + academic* sources into a single thematic narrative;
2. proposes an explicit, reusable seven-dimension scoring ontology for healthcare AI compliance;
3. delivers an open interactive interface (map, radar, heatmap, trends, use-case readiness) rather than a static PDF-only review;
4. documents reproducibility (JSON schema, code listings, environment) so that scores and charts can be regenerated from the same sources;
5. maintains HITL accountability: automated assistance must not replace expert judgment on regulatory interpretation.

Identified gaps that remain open—and motivate Chapter 4—include limited labeled corpora for clause-level NLP classification, incomplete device-count transparency in some jurisdictions, and the absence of longitudinal score versioning tied to legislative change events.

2.11 Deep Dive: International Harmonization Instruments

Beyond national law, several international instruments shape practical compliance strategies for multinational AI medical products.

2.11.1 IMDRF as Conceptual Backbone

The IMDRF SaMD risk framework remains the most widely referenced conceptual backbone for classifying software medical functions [11]. Its two-axis logic (significance of information $\times$ healthcare situation) is intentionally technology-agnostic, which helped regulators absorb AI/ML without inventing entirely new class taxonomies. However, IMDRF guidance is not self-executing law: national transposition, guidance density, and agency capacity determine real-world effects. Comparative dashboards must therefore distinguish *adoption of IMDRF vocabulary* from *operational maturity of AI-specific review*.

2.11.2 GMLP and Cross-Agency Soft Law

The 2021 GMLP principles jointly issued by FDA, Health Canada, and MHRA illustrate soft-law harmonization [19]. They emphasize multi-disciplinary expertise, good software engineering, representative clinical data, reference standards, model design for human interpretability where appropriate, and monitoring. Because GMLP is principle-based rather than checklist-prescriptive, jurisdictions can claim alignment while differing in evidentiary thresholds. For this project, GMLP informs the clinical-validation and transparency theme rubrics.

2.11.3 WHO Ethics Guidance as Normative Reference

WHO's ethics and governance guidance for AI in health provides a globally portable normative vocabulary—autonomy, well-being, transparency, responsibility, inclusiveness, sustainability [24]. It is particularly valuable for emerging regulators that have data-protection statutes but thin AI-device doctrine. The ethics theme in our ontology therefore rewards not only the existence of national AI ethics documents but also signals of institutionalization (committees, impact assessments, procurement rules).

2.11.4 OECD and NIST Risk Management

OECD.AI observatory materials and the NIST AI RMF offer voluntary risk-management scaffolding used by ministries and enterprises [22], [29]. They rarely replace SaMD law, yet they increasingly appear in governance narratives of advanced jurisdictions that lack horizontal AI Acts. Scoring must avoid double-counting voluntary frameworks as equivalent to binding high-risk duties under the EU AI Act [14].

2.12 Clinical Domains as Regulatory Stress Tests

Regulatory theory becomes concrete when mapped onto clinical workflows.

2.12.1 Radiology and Imaging

Imaging AI is the most mature commercial SaMD segment. Literature repeatedly flags external validation gaps and subgroup performance issues [3], [21]. Compliance emphasis: multi-site validation, intended-use precision, clinician-facing disclosures, and post-deployment monitoring when integrated into PACS/RIS.

2.12.2 Digital Pathology

Whole-slide imaging introduces scanner variability and labeling cost as regulatory-relevant quality factors. Data governance intersects with biobank ethics. Lifecycle surveillance matters when models are fine-tuned on institutional slides.

2.12.3 Genomics and Precision Medicine

Genomic AI intensifies privacy, representativeness, and cross-border transfer constraints. Fairness failures can entrench ancestry-correlated disparities. Theme weights in the dashboard therefore elevate privacy for genomics readiness [5].

2.12.4 Drug Discovery and Development Support

AI used upstream of clinical care may escape classic SaMD definitions yet still affect evidence packages. Documentation, auditability, and liability assumptions become central even without a bedside device label [28].

2.12.5 Hospital CDS and Public Health

Embedded CDS requires Total Product Lifecycle thinking [17], [27]. Public-health AI expands the rights-holder set from patients to populations, raising accountability and contestability requirements [24].

2.12.6 Foundation and Generative Models

Foundation models challenge intended-use specificity, update cadence, and evaluation protocols assumed by classical SaMD pathways. Literature and regulators are still converging; our state-of-the-art treats this as an open frontier rather than a settled comparative axis.

2.13 Systematic Review Methods in Regulatory Research

Phase 1 referenced PRISMA-inspired workflows for systematic evidence synthesis. While this PFE is a structured thematic review rather than a full meta-analysis, several PRISMA disciplines transfer: protocol-like search documentation, inclusion criteria, source provenance, and transparent limitations. Automation via NLP is positioned as an accelerator for screening and clause retrieval, not as an unsupervised substitute for citation-worthy claims. HITL checkpoints are therefore methodological requirements, not optional UX flourishes.

2.14 Industrial Solutions and Benchmarks

Commercial regulatory-intelligence platforms (pharma/medtech) provide curated alerts and jurisdiction matrices, but typically under proprietary licenses incompatible with open academic reproducibility. Open policy observatories provide breadth without healthcare-AI theme ontologies tuned for SaMD. Academic papers often compare two regions deeply rather than twenty shallowly [15]. This project occupies the middle ground: open data, multi-region coverage, healthcare-specific themes, and interactive exploration—accepting that depth per country is curated-summary rather than doctrinal treatise.

2.15 Critical Gaps Summarized

The literature and policy corpus jointly reveal five gaps this work partially addresses:

1. missing open, versioned scorecards linking law-on-the-books to capacity signals;
2. weak interoperability between ethics principles and measurable compliance indicators [25], [26];
3. under-specified post-market metrics for AI-specific harms [27];
4. liability doctrines lagging deployment reality [28];
5. insufficient gold-standard corpora for supervised regulatory NLP.

These gaps motivate the scoring ontology, dashboard design, and NLP scale-up path detailed next.

2.16 Additional Related Work on Dashboards and Policy Analytics

Interactive policy dashboards are increasingly used in public health and digital-governance research to communicate multi-indicator comparisons. Best practices include disclosing indicator definitions, providing downloadable underlying data, and avoiding overprecise visual encodings for ordinal expert scores. The dashboard developed in this work follows these practices by exposing theme definitions, CSV export, and narrative caveats alongside charts. Unlike purely descriptive observatories, it couples the interface to an explicit healthcare-AI compliance ontology, which is the main methodological differentiator relative to generic AI-policy trackers.

2.17 Extended Discussion: From Principles to Enforceable Controls

A central lesson of the 2018–2026 literature is that ethical principles proliferate faster than enforceable controls [23], [25]. Healthcare intensifies the problem because patient harm is tangible and asymmetric: false reassurance from an opaque model can delay care, while overly conservative regulation can deny beneficial tools to underserved clinics. Comparative research must therefore track not only whether a jurisdiction has an ethics charter, but whether procurement, audit, incident reporting, and liability pathways translate principles into operational duties [24], [26].

In the European Union, the combination of GDPR, MDR, and the AI Act is the clearest attempt to bind high-risk healthcare AI to documentation, data governance, human oversight, and ongoing conformity [10], [14]. In the United States, pathway maturity and GMLP soft law enable rapid authorization while leaving ethics and transparency comparatively less horizontally mandated [3], [4], [19]. East Asian systems demonstrate that industrial policy and algorithm-specific rules can accelerate market entry while raising localization and dual-use governance questions [12], [13]. Emerging jurisdictions often legislate comprehensive privacy first—a necessary but insufficient condition for safe clinical AI scale-up.

For this project, these observations justify a multi-theme ontology rather than a single "AI readiness" index. They also justify interactive exploration: stakeholders disagree about which theme should dominate, and the dashboard allows that disagreement to be inspected rather than hidden. Finally, they justify HITL NLP: legal meaning is context-sensitive, and automated classifiers should propose candidate clauses for expert review instead of silently rewriting compliance truth.

2.18 Notes on Evidence Quality in Regulatory Science

Regulatory science for AI medical devices is itself an evolving field. Studies of FDA clearances highlight limitations in validation reporting [3]; demographic bias studies show how population mismatch becomes a safety issue [20], [21]; lifecycle papers argue that post-market systems must change [16], [17]. The state of the art is therefore not only a catalog of laws but a catalog of *evaluation deficits*. Any compliance dashboard that ignores evidence-quality debates risks celebrating paperwork over patient outcomes. This PFE keeps validation and post-market themes first-class for that reason.

2.19 Foundational Learning Systems and Representation Learning

Although this project is not a predictive modeling PFE, its NLP roadmap and dashboard analytics inherit concepts from contemporary machine learning. Representation learning and attention-based architectures underpin modern document encoders used for clause retrieval [30]. Broader surveys of deep learning situate why opaque high-capacity models create governance pressure in clinical settings [31]. Healthcare-specific deep learning reviews further document imaging and multimodal progress that regulators must accommodate without freezing innovation [32], [33].

For compliance analytics, the relevant transfer is methodological rather than architectural:

curated labels, transparent metrics, and human oversight remain necessary even when models are strong. The state of the art therefore cautions against treating either “more AI” or “more paperwork” as sufficient; both evidence quality and enforceable controls matter.

2.20 Comparative Tables and Synthesis Conventions

Good comparative reviews in digital health increasingly use structured tables to expose incommensurable categories (device class, privacy regime, AI-specific duties). This chapter follows that convention: risk-class tables, theme syntheses, and explicit gap lists are preferred over narrative-only summaries. The dashboard later operationalizes the same convention interactively, so that readers can re-order and filter the synthesis rather than accept a single authorial ranking.

2.21 Positioning Summary for Downstream Chapters

In sum, prior work provides rich jurisdiction-specific guidance and strong ethics discourse, but comparatively weak open multi-theme scorecards linking law, capacity, and clinical use-case constraints. Chapter 4 converts that positioning into a reproducible ontology, dataset schema, and interactive system; Chapter 5 reports what the resulting indicators show across twenty jurisdictions.

3. Project Context and Scope

3.1 Host Setting and Organizational Context

This Final Year Project is hosted and supervised at aivancity under the academic guidance of Prof. Anuradha Kar. Unlike a classical industry internship embedded in a commercial product team, the mission is an academic applied-research project: the "host" is the school itself, and the deliverables are a positioned literature synthesis, a curated multi-country compliance dataset, and an interactive analytics dashboard for AI healthcare regulation.

aivancity's AI & Data Science programme emphasizes projects that connect technical implementation with societal and regulatory stakes. That institutional setting shapes both the problem choice and the acceptance criteria: the work must be scientifically positioned, reproducible, and communicable to academic and professional audiences, not merely a demo application.

3.2 Team, Supervision, and Collaboration Mode

Day-to-day scientific supervision is provided by the academic tutor. Feedback cycles focus on (i) clarity of the research question, (ii) quality of thematic synthesis, (iii) transparency of the scoring protocol, and (iv) honesty about limitations. There is no separate industry internship supervisor for this project; cover-page fields therefore record the internship supervisor as not applicable, while retaining aivancity as host company in the template sense of the hosting organization.

Collaboration is asynchronous and document-centric: drafts of chapters, dataset versions, and dashboard builds are reviewed against methodological criteria rather than sprint demos alone. This matches the guideline expectation that context sections describe real roles and constraints rather than promotional narratives.

3.3 Role and Responsibilities

The author combines research-analyst and full-stack data/AI engineering responsibilities:

- design the review protocol and thematic coding frame;
- curate jurisdictional profiles and theme scores with documented sources;
- implement data schemas and dashboard features (Streamlit analytics application);
- generate reproducible report artifacts (LaTeX/PDF, literature exports, figure regeneration scripts);
- iterate with the academic supervisor on scientific positioning and writing quality;
- publish and maintain the open repository used for reproducibility audits.

Expected deliverables include: Phase 1 topic framing; literature review; compliance dataset; interactive dashboard (`https://github.com/remi7025/AI-Healthcare-Compliance-Dashbo` and this Final Project Report.

3.4 Mission Scope and Timeline

The work progressed through successive milestones aligned with the programme schedule:

- Phase 1 topic framing (comparative review + dashboard concept);
- introduction and state-of-the-art drafts;
- methodology and preliminary results (dataset scores + dashboard views);
- full draft for supervisor feedback;
- final report, sources, and code package for platform submission.

Within that timeline, priority was given to a correct curated MVP: open JSON as system of record, interactive comparison views, and a written report that can be audited against primary sources. Fully supervised NLP classification and live legislative scraping were intentionally deferred to the roadmap rather than claimed as finished products.

3.5 Data Accessed and Information Assets

The project primarily uses *public* regulatory and scientific information:

- agency guidance and legislation (FDA, EU AI Act/MDR/GDPR, MHRA, NMPA, PMDA, WHO, OECD, NIST);
- peer-reviewed literature via PubMed, Scopus, IEEE Xplore, and related databases;
- curated structured records stored in `compliance_dataset.json` (twenty country objects, theme scores, trends, references);
- secondary comparative studies of device authorizations and ethics surveys used for calibration [3], [15], [23].

Data volume is modest in the machine-learning sense (tens of jurisdictional records rather than millions of patients) but high in interpretive density: each country object encodes narrative fields that must remain consistent with citations. Quality control therefore emphasizes provenance and rubric adherence more than big-data pipelines.

No identifiable patient-level EHR data are processed in the current dashboard release. Any future integration of de-identified clinical records for compliance auditing would require a separate DPIA, legal basis, and written authorization before inclusion in a disseminated report.

3.6 Legal, Ethical, and Confidentiality Constraints

- **GDPR / privacy:** literature and public law texts are processed; no special-category patient data in v1.

- **Confidentiality:** proprietary materials, if introduced later, must be redacted or confined to a confidential report variant; the present report is intended as a non-confidential academic version.
- **Not legal advice:** comparative scores are research indicators for comparative analysis, not legal opinions.
- **Generative AI:** any assistant use is declared in the appendix; analyses and conclusions remain the author's responsibility.
- **Dissemination:** authorization is required before public release if confidentiality constraints apply; open code publication covers only public-source analytics.
- **Conflicts of interest:** none declared; scores are not commissioned by any national regulator or vendor.

3.7 Technical and Resource Constraints

Constraints shaping design choices include: reliance on publicly documented device counts (heterogeneous reporting); English-dominant source coverage (potential language bias); local execution without paid regulatory-intelligence APIs; and project-time bounds that prioritize a robust curated MVP over fully supervised NLP classifiers. Hardware needs remain laptop-class for the Streamlit application; GPU resources are optional and reserved for future encoder fine-tuning.

These constraints explain several methodological decisions: expert-curated scores rather than fully automated extraction; JSON rather than a multi-tenant database; and dual emphasis on tables (precision) and charts (pattern discovery). The same constraints are revisited in the limitations section of Chapter 6.

3.8 Risk Register for the Mission

Key project risks and mitigations include:

- **Source obsolescence:** mitigated by dataset `version`/`last_updated` fields and changelog discipline;
- **Misinterpretation of law:** mitigated by non-legal-advice caveats and human-in-the-loop score governance;
- **Scope creep into clinical model building:** mitigated by explicit non-goals in Chapter 1;
- **UI maintenance cost:** mitigated by a shared JSON system of record;
- **Citation drift:** mitigated through a single BibTeX / biblatex IEEE configuration shared by report and literature views.

3.9 Budgetary and Organizational Non-Goals

The mission does not include paid access to commercial regulatory databases, clinical trial sponsorship, or deployment inside a hospital EHR. Budget is effectively zero beyond personal compute and open tooling. Organizationally, the project does not replace institutional compliance functions; it produces a research artifact that those functions could later adapt.

3.10 Supervision and Acceptance Criteria

Scientific guidance is provided by Prof. Anuradha Kar. Acceptance for the written report emphasizes: (1) a clear research question; (2) thematic rather than enumerative state of the art; (3) reproducible scoring and dashboard launch instructions; (4) critical discussion of validity threats; and (5) alignment with aivancity PFE structure and IEEE citation practice. Cosmetic dashboard features are secondary to those criteria.

3.11 Bridge to Methodology

Given this academic-host setting, public-data constraint, and reproducibility mandate, Chapter 4 details the end-to-end architecture, search protocol, seven-theme ontology, evaluation metrics for the analytics artifact, and the NLP scale-up path that remains intentionally incomplete in the present release.

4. Methodology and Technical Approach

This chapter is the scientific core of the report. It is written to be reproducible: another researcher should be able to re-implement the scoring workflow, regenerate tables, and relaunch the dashboard from the provided schema and code.

4.1 General Solution Architecture

Figure 4.1 summarizes the end-to-end pipeline from sources to decision-support views.

Stage	**Description**
Sources	Agency documents and peer-reviewed papers
Acquisition	Structured search protocol and provenance logging
Curation / NLP	Theme coding with optional embeddings for scale-up
JSON dataset	Twenty countries, seven themes, trends, references
Dashboard	Map, radar, heatmaps, trends, use-case readiness
Exports	CSV tables, PDF reports, presentation slides

Figure 4.1: End-to-end architecture of the compliance analytics system. Source: author.

Functional blocks:

1. **Acquisition:** structured search across bibliographic databases and official repositories;
2. **Curation:** thematic coding into seven compliance dimensions with narrative fields;
3. **Optional NLP assist:** preprocessing, embeddings, and draft classification for future scale-up;
4. **Storage:** version-controlled JSON as the system of record;
5. **Presentation:** Streamlit analytics prototype and React BI-style web app;
6. **Reporting:** LaTeX/PDF generation and literature rendering inside the UI.

4.2 Literature Search and Inclusion Protocol

4.2.1 Databases and Regulatory Sources

Search covered PubMed, Scopus, IEEE Xplore, Google Scholar, and SSRN, complemented by FDA, EMA, MHRA, NMPA, WHO, OECD, and national AI strategy portals. Inclusion emphasized 2018–2026 publications addressing AI regulation, compliance, or ethics in healthcare, plus foundational pre-2018 frameworks still in force (e.g., IMDRF 2014, HIPAA).

4.2.2 Query Design

Representative query families included: “AI healthcare regulation”, “SaMD compliance”, “health data governance”, “algorithmic transparency medicine”, “EU AI Act medical device”, and “post-market surveillance AI”. Snowballing from key reviews (e.g., device-approval comparisons, ethics surveys) complemented keyword search [15], [23].

4.2.3 Screening Logic

Titles/abstracts were screened for healthcare + AI + regulatory/ethical relevance. Full texts informed theme narratives. Grey literature (agency PDFs) was admitted when it constituted primary law or official guidance. Informal encyclopedias and blogs may inform discovery but do not justify score cells or bibliographic claims.

4.3 Seven-Theme Scoring Ontology

Table 4.1: Compliance themes used for jurisdictional scoring

ID	Theme
1	Data Privacy & Governance
2	Clinical Validation & Safety
3	Regulatory Approval Process
4	Algorithmic Transparency
5	Ethical Considerations
6	Post-Market Surveillance
7	Liability & Accountability

Each theme is scored on an integer scale from 1 (minimal formalization / capacity) to 10 (mature, AI-aware, enforceable framework). Scores are comparative indicators combining: existence of AI-relevant rules; enforceability; agency capacity signals; and alignment with international good practice (IMDRF, GMLP, WHO ethics). Higher is not automatically "better for innovation"; some high scores reflect stringent precaution.

4.3.1 Calibration Rules

- Prefer primary legal texts over secondary commentary when assigning scores;
- Separate "law on the books" from evident implementation capacity when narratives disagree;
- Record device-approval counts as a complementary throughput metric, not a quality proxy alone;
- Document uncertainty in country narrative fields (`challenges`, `notable_developments`).

4.4 Dataset Schema and Curation Workflow

The canonical file `data/compliance_dataset.json` contains:

- `metadata` — title, version, theme list, sources;
- `countries` — array of jurisdictional objects;
- `global_trends` — structured trend cards;
- `key_references` — bibliographic pointers.

Country objects include identifiers (`country`, `iso_code`, `region`), institutional fields (`regulatory_body`, privacy law, AI-specific regulation, medical-device framework), narrative dimensions aligned to themes, `key_legislations`, `maturity_level`, `year_first_ai_regulation`, `num_ai_devices_approved`, and `themes_scores`.

$$S_c = \frac{1}{7}\sum_{t=1}^{7} s_{c,t} \tag{4.1}$$

where $s_{c,t} \in [1, 10]$ is the score of country c on theme t, and S_c is the unweighted composite used for choropleth coloring unless a user applies filters.

4.5 NLP Pipeline Design (Scale-Up Path)

Phase 1 proposed an assembly-line NLP approach. The draft methodology retained for reproducibility is:

4.5.1 Data Collection and Ingestion

Sources include official HTML/PDF portals and bibliographic APIs. Acquisition scripts should log URL, access date, and checksum. Update frequency for a production system would be monthly for agency newsrooms and event-driven for major legislative acts.

4.5.2 Preprocessing

1. PDF/HTML text extraction;
2. language detection and optional translation for indexing;
3. normalization (Unicode, hyphenation repair);
4. segmentation into articles/sections/clauses;
5. metadata tagging (jurisdiction, instrument type, date).

4.5.3 Feature Extraction

Lexical baselines (TF–IDF) provide interpretable sparse features. Dense embeddings (e.g., domain-adapted BERT) capture semantic similarity between clauses on "clinical validation" or "human oversight." Equation (4.2) denotes sentence embedding $\mathbf{e}_i$ for segment i:

$$\mathbf{e}_i = f_\theta(x_i) \in \mathbb{R}^d \tag{4.2}$$

with encoder parameters θ frozen or fine-tuned depending on labeled data availability.

4.5.4 Classification and HITL

Draft labels map segments to themes and coarse risk cues. Human-in-the-loop review is mandatory before scores change: NLP accelerates scanning; experts remain accountable for regulatory meaning.

4.6 Dashboard Implementation

4.6.1 Technology Stack

Table 4.2: Implementation stack

Component	Technology
Analytics prototype	Streamlit + Plotly + Pandas
Presentation web app	React + Recharts + react-simple-maps
Data store	Versioned JSON
Report generation	LaTeX, fpdf2, python-pptx
Runtime	Python 3.9+, Node.js (web)

4.6.2 Information Architecture

Primary views: World Map; Country Comparison; Theme Analysis; Global Trends; Country Details; AI Use Cases & Trends; Literature Review. Filters include region, maturity, and minimum theme score. CSV export supports external audit of the exact filtered subset.

4.6.3 Use-Case Readiness Model

For clinical domains u (radiology, pathology, genomics, drug discovery), a derived readiness score uses theme weights $w_{u,t}$:

$$R_{c,u} = \sum_{t=1}^{7} w_{u,t}\, s_{c,t} \quad \text{with} \quad \sum_{t} w_{u,t} = 1,\ w_{u,t} \geq 0. \tag{4.3}$$

Radiology emphasizes clinical validation and post-market surveillance; genomics emphasizes privacy; drug discovery emphasizes approval process and transparency. Weights are disclosed in the UI to avoid black-box rankings.

4.6.4 Evaluation Metrics for the Software Artifact

Unlike a single predictive model, evaluation combines:

- **Coverage:** 20/20 countries with complete theme vectors;
- **Consistency:** schema validation and non-null critical fields;
- **Usability:** task-based walkthroughs (compare two countries; identify lowest liability scores);
- **Reproducibility:** one-command local launch;
- **Traceability:** each narrative linked to legislation lists and bibliography.

4.7 Baselines and Alternatives Considered

Methodological alternatives rejected or deferred:

- **Pure qualitative review:** insufficiently actionable for comparative dashboards;
- **Fully automated scraping without curation:** high factual risk for legal text;

- **Single composite index without themes:** hides policy trade-offs (e.g., high approvals vs weak transparency);
- **Closed commercial data only:** conflicts with academic reproducibility.

The chosen hybrid—expert curation + interactive analytics + NLP-ready schema—optimizes for scientific reproducibility and practical stakeholder use.

4.8 Environment, Versioning, and Reproducibility

Code and data for the interactive dashboard are published at `https://github.com/remi7025/AI-Healthcare-Compliance-Dashboard` (`app.py`, `requirements.txt`, `literature_revi` and `data/compliance_dataset.json`). Dependencies are pinned via `requirements.txt`. Dataset `version` and `last_updated` fields support audit trails. Listing 4.1 shows the canonical Streamlit launch command.

```
pip install -r requirements.txt
python -m streamlit run app.py
# open http://localhost:8501
```

Listing 4.1: Launching the Streamlit compliance dashboard

4.9 Detailed Scoring Rubric by Theme

To improve reproducibility, Table 4.3 summarizes ordinal anchors used during curation. Intermediate scores (e.g., 5–6) indicate partial AI-specificity or uneven enforcement capacity.

4.10 Worked Example of Score Assignment

Consider a stylized comparison between a jurisdiction with a binding high-risk AI law layered on MDR/GDPR versus one with high device throughput but sectoral privacy and voluntary ethics toolkits. The first may score higher on transparency, ethics, and privacy even if the second reports more device authorizations. Equation (4.1) deliberately uses an unweighted mean so that no single theme dominates; users who need policy-weighted views can recompute S_c externally from exported CSV.

4.11 Streamlit Application Structure

The Streamlit prototype loads JSON at startup, normalizes records into a Pandas DataFrame, and renders:

- sidebar filters (region, maturity, minimum theme thresholds);
- executive KPI cards (top/bottom composites, largest theme gaps);
- choropleth mapping of S_c;
- pairwise comparison bars and radar charts;
- theme heatmaps and distributions;

Table 4.3: Simplified ordinal anchors for theme scores (1–10)

Theme	Low (1–3)	High (8–10)
Data privacy	Minimal health-data rules; weak enforcement	Comprehensive law + health-sensitive rules + transfer tools
Clinical validation	No AI-aware evidence expectations	Clear AI evidence guidance, bias/subgroup expectations, RWE pathways
Approval process	General device law only, unclear AI path	Explicit AI/SaMD pathways, sandboxes, predictable review
Transparency	No documentation/logging expectations	Binding technical docs, logging, user notices, oversight duties
Ethics	Absent or purely rhetorical principles	Institutionalized ethics tools linked to procurement/oversight
Post-market	Paper adverse-event systems only	Lifecycle monitoring, update governance, AI-aware vigilance
Liability	No AI-relevant clarity	Explicit reforms or mature case guidance allocating responsibility

- trend cards with adoption metadata;
- country narrative accordion pages;
- derived use-case readiness charts;
- embedded literature markdown browser.

Custom CSS provides a presentation-ready layout for stakeholder demos. Because all data are local, the demo requires no API keys—an intentional constraint for offline and reproducible deployment.

4.12 React Presentation Layer

A React + Vite application mirrors the analytical pages with a Power-BI-inspired layout (header, slicers, KPI tiles, multi-visual pages). Libraries include Recharts for plots and react-simple-maps for geospatial views. A sync script copies the JSON dataset into the web public folder to keep both UIs aligned. This dual-implementation strategy separates rapid scientific iteration (Python) from polished stakeholder storytelling (TypeScript/React).

4.13 Quality Assurance Protocol

Before cutting a dataset version:

1. schema validation (required keys, score ranges, ISO codes);
2. cross-check legislation lists against narrative claims;
3. spot-audit of extreme scores (10s and 1–2s) against primary sources;
4. regenerate LaTeX score tables and visually inspect ranking shifts;
5. smoke-test dashboard filters and CSV export round-trip.

4.14 Threats to Methodological Validity

Construct validity threats include theme overlap (ethics vs transparency) and the difficulty of encoding enforcement capacity. Internal validity threats include curator drift over time. External validity threats include English-source bias and EU aggregation. The mitigation strategy is transparency: publish rubrics, narratives, and version metadata so peers can contest and improve scores.

4.15 Hyperparameters and Configuration Surfaces

Although the core artifact is not a trained classifier in this draft, configuration surfaces exist: theme weights for use-case readiness (Equation (4.3)), filter thresholds in the UI, and embedding model choice for future NLP. These should be treated analogously to ML hyperparameters—logged, justified, and held fixed when comparing dashboard releases.

4.16 Data Provenance and Citation Discipline

Every country record stores legislation titles and narrative fields that should be traceable to agency documents or peer-reviewed analyses. In practice, curators keep a working bibliography (BibTeX) and paste short provenance notes into `notable_developments` when a score changes. This discipline reduces silent drift between the dashboard and the written report.

When secondary sources disagree, the protocol prefers: (1) primary legal text; (2) official agency guidance; (3) peer-reviewed comparative studies; (4) reputable policy observatories. Blog commentary and encyclopedia pages may inform discovery but do not justify score cells.

4.17 Interface Wireframe Logic

The information architecture follows progressive disclosure. Level 1 presents a world overview and KPI strip. Level 2 supports filtered comparison. Level 3 opens country narratives and legislation lists. Level 4 surfaces literature and methodological caveats. This mirrors how compliance officers actually work: start from a map of exposure, then drill into obligations.

4.18 Ethics of Comparative Scoring

Publishing comparative scores can create reputational effects for countries. The report therefore repeats that scores are research instruments, not rankings for investment or diplomacy. Emerging jurisdictions may score lower because documentation is scarce in English or because AI-SaMD pathways are genuinely early—two different situations that narratives must disentangle. Capacity-building recommendations in the conclusion are intended to counteract extractive “naming and shaming” readings of the dashboard.

4.19 Training and Onboarding Notes

A new contributor to the dataset should: read the rubric table; shadow one country update end-to-end; propose scores with written justifications; and only then merge after supervisor or peer review. This onboarding mirrors HITL principles used in the NLP pathway and protects longitudinal consistency.

4.20 Mapping Phase 1 Claims to Implemented Features

Phase 1 promised a multi-level presentation from global map to compliance alerts, real-time monitoring, and HITL verification. Table 4.4 maps those claims to the current draft status.

This traceability matters for scientific integrity: the draft prioritizes correctness and reproducibility of curated analytics before fully automated monitoring.

Table 4.4: Phase 1 ambitions versus draft implementation status

Phase 1 claim	Status	Notes
Comparative literature synthesis	Delivered	Thematic SotA + bibliography
Multi-country structured dataset	Delivered	20 jurisdictions, 7 themes
Interactive dashboard (map/compare)	Delivered	Streamlit + React
NLP classification pipeline	Designed	Schema-ready; full training future
Real-time legislative monitoring	Roadmap	Versioned curation first
HITL verification	Delivered (process)	Manual score governance

4.21 Resource Estimation for Full NLP Scale-Up

A minimal viable supervised classifier would require on the order of several thousand annotated clause segments across themes, dual annotation for agreement measurement, and a validation split by jurisdiction to test geographic generalization. Compute needs remain modest relative to foundation-model pretraining: a single GPU workstation suffices for fine-tuning an encoder on clause classification. The larger cost is expert annotation time—precisely why HITL dashboards that accelerate human review are valuable even before model metrics peak.

4.22 Final Methodological Checklist

Before any public release of a new dataset version, confirm: all theme scores in $1..10$; no empty required narratives for Advanced jurisdictions; bibliography keys cited in text resolve; dashboard launches from a clean virtual environment; and the LaTeX report regenerates without undefined references. This checklist operationalizes reproducibility as a recurring practice rather than a one-time appendix claim.

4.23 Closing Bridge to Results

The methodology above converts regulatory complexity into a measurable, explorable artifact. Chapter 5 reports what the curated scores and dashboard views reveal: who leads on which themes, where gaps concentrate, and how clinical use-case lenses change the interpretation of the same underlying data.

4.24 Data Collection and Ingestion Details

Sources enter the pipeline through a lightweight ETL pattern. Agency PDFs and HTML guidance pages are logged with URL, access date, and jurisdiction tags. Bibliographic hits are exported to BibTeX and deduplicated by DOI/title. Country records are updated in JSON rather than overwritten silently: when a theme score changes, the curator records a short justification in narrative fields. Update frequency is event-driven (new law or guidance) rather than continuous crawling, which matches the correctness-first constraint of Chapter 3.

4.25 Preprocessing for Future NLP Assist

For the planned NLP scale-up, documents would be converted to plain text, segmented into clauses or paragraphs, normalized for whitespace, and embedded with a transformer encoder [30], [34], [35]. Train/validation/test splits should be stratified by jurisdiction and theme to avoid leaking country-specific boilerplate into metrics. Libraries envisaged for that path include pandas for tabular joins, scikit-learn for baseline classifiers, and Hugging Face Transformers for encoders. Foundational references on deep learning and AI systems provide the conceptual background for these choices [31], [36], [37]. In the present release, these steps remain design-complete rather than fully trained—a deliberate honesty boundary.

4.26 Modeling Choices for the Analytics Artifact

The "model" of this PFE is primarily an expert scoring model plus interactive analytics, not a neural predictor. Ordinal theme scores are the parameters; the composite S_c and use-case readiness scores are derived quantities. Alternative modeling choices considered and rejected include: (i) a single opaque readiness index; (ii) fully automated scraping without curation; (iii) vendor-locked BI platforms that hinder reproducibility. The chosen hybrid maximizes auditability for an academic jury while remaining usable for professional briefing [38], [39].

4.27 Training, Infrastructure, and Cost Profile

Infrastructure is intentionally minimal: a local Python environment, Streamlit for UI, and MiKTeX/TeX Live for report compilation. No cloud GPU spend was required for the delivered MVP. Estimated cost is dominated by researcher time for curation and writing. This profile is a feature for academic reproducibility: any examiner can clone the repository and relaunch without institutional cloud credentials.

4.28 Evaluation Metrics for the Delivered System

Because the primary artifact is curated comparative analytics, evaluation uses research-software metrics rather than clinical AUC alone:

- **Coverage:** twenty jurisdictions $\times$ seven themes with non-empty narratives for advanced markets;
- **Consistency:** schema validation and score-range checks;
- **Reproducibility:** clean-environment dashboard launch and regenerated tables/figures;

- **Comparative utility:** ability to surface theme gaps (e.g., privacy vs liability) against literature expectations;
- **Future NLP metrics:** planned precision/recall/F1 on clause classification once labels mature.

Baselines for interpretation are literature-reported authorization patterns and regime characterizations [3], [14], [15], not a competing commercial dashboard score.

4.29 Threats to Methodological Validity

Construct validity threats include imperfect mapping from ordinal scores to real enforcement. Internal validity threats include curator bias. External validity threats include English-source skew and EU-as-aggregate smoothing. Conclusion validity threats include over-reading device counts. Each threat is mitigated partially (rubrics, peer review, caveats) and discussed again in Chapter 5.

5. Results and Analysis

5.1 Quantitative Overview

Table 5.1 ranks jurisdictions by unweighted composite score S_c (Equation (4.1)). Table 5.2 summarizes global theme means and ranges. The full seven-theme matrix appears in Table 5.3. Visual summaries generated from the open dataset published with the Streamlit dashboard[1] are shown in Figures 5.1–5.7.

Table 5.1: Jurisdictions ranked by composite compliance score S_c

Country	Region	Maturity	S_c	Devices
European Union	Europe	Advanced	9.14	600
Germany	Europe	Advanced	8.71	180
United States	North America	Advanced	7.57	950
United Kingdom	Europe	Advanced	7.14	200
Canada	North America	Advanced	7.14	120
Japan	Asia	Advanced	7.00	150
Singapore	Asia	Advanced	6.86	60
Switzerland	Europe	Advanced	6.86	90
China	Asia	Advanced	6.71	300
Australia	Oceania	Moderate	6.57	80
South Korea	Asia	Advanced	6.57	200
Israel	Middle East	Advanced	6.29	100
Brazil	South America	Developing	5.29	40
UAE	Middle East	Emerging	4.57	35
Saudi Arabia	Middle East	Emerging	4.43	25
India	Asia	Developing	4.29	30
Thailand	Asia	Developing	3.86	15
South Africa	Africa	Emerging	3.57	10
Kenya	Africa	Early	2.71	3
Nigeria	Africa	Early	2.57	5

[1] https://github.com/remi7025/AI-Healthcare-Compliance-Dashboard

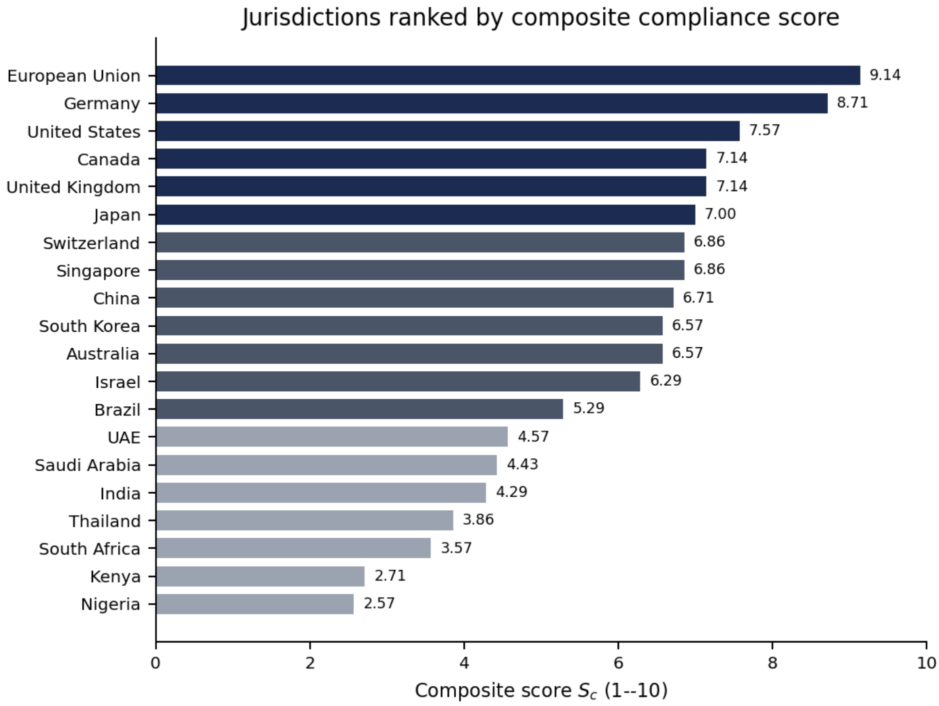

Figure 5.1: Composite score ranking across 20 jurisdictions (source: curated dataset).

Table 5.2: Global theme score statistics across 20 jurisdictions

Theme	**Mean**	**Min**	**Max**
Data privacy	6.70	4	10
Clinical validation	6.20	2	9
Approval process	6.30	3	9
Transparency	5.60	2	9
Ethics	5.95	3	10
Post-market	5.75	2	9
Liability	4.75	2	8

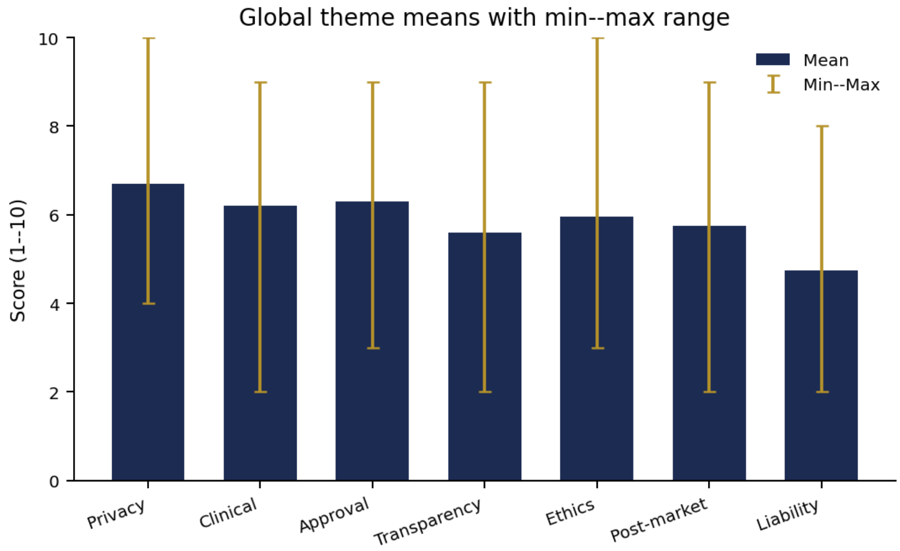

Figure 5.2: Global theme means with min–max ranges.

Table 5.3: Country compliance scores (1–10) across seven themes

Country	Region	Maturity	Priv.	Clin.	Appr.	Trans.	Eth.	Post	Liab.
European Union	Europe	Advanced	10	9	9	9	10	9	8
Germany	Europe	Advanced	10	9	9	8	9	9	7
United States	North America	Advanced	8	9	9	6	6	8	7
United Kingdom	Europe	Advanced	8	8	7	7	7	7	6
Canada	North America	Advanced	7	8	8	7	7	7	6
Japan	Asia	Advanced	7	8	8	6	7	8	5
Singapore	Asia	Advanced	7	7	7	8	8	6	5
Switzerland	Europe	Advanced	8	8	8	6	6	7	5
China	Asia	Advanced	7	7	7	7	6	7	6
Australia	Oceania	Moderate	7	7	7	6	7	7	5
South Korea	Asia	Advanced	7	7	8	6	6	7	5
Israel	Middle East	Advanced	6	8	7	6	6	6	5
Brazil	South America	Developing	7	5	5	5	5	5	5
UAE	Middle East	Emerging	5	5	5	5	5	4	3
Saudi Arabia	Middle East	Emerging	5	4	5	5	5	4	3
India	Asia	Developing	5	4	4	4	5	4	4
Thailand	Asia	Developing	5	4	4	4	4	3	3
South Africa	Africa	Emerging	6	3	3	3	4	3	3
Kenya	Africa	Early	5	2	3	2	3	2	2
Nigeria	Africa	Early	4	2	3	2	3	2	2

Table 5.4: Geographic coverage of the compliance dataset (20 jurisdictions)

Region	**Jurisdictions**
Africa	South Africa, Nigeria, Kenya
Asia	China, India, Japan, South Korea, Singapore, Thailand
Europe	European Union, United Kingdom, Switzerland, Germany
Middle East	Saudi Arabia, Israel, UAE
North America	United States, Canada
Oceania	Australia
South America	Brazil

Table 5.5: Count of jurisdictions by regulatory maturity label

Maturity	**Count**
Advanced	11
Moderate	1
Developing	3
Emerging	3
Early	2

Table 5.6: Regional mean theme scores (1–10) derived from the curated dataset

Region	**Priv.**	**Clin.**	**Appr.**	**Trans.**	**Eth.**	**Post**	**Liab.**
Africa	5.0	2.3	3.0	2.3	3.3	2.3	2.3
Asia	6.3	6.2	6.3	5.8	6.0	5.8	4.7
Europe	9.0	8.5	8.2	7.5	8.0	8.0	6.5
Middle East	5.3	5.7	5.7	5.3	5.3	4.7	3.7
North America	7.5	8.5	8.5	6.5	6.5	7.5	6.5
Oceania	7.0	7.0	7.0	6.0	7.0	7.0	5.0
South America	7.0	5.0	5.0	5.0	5.0	5.0	5.0

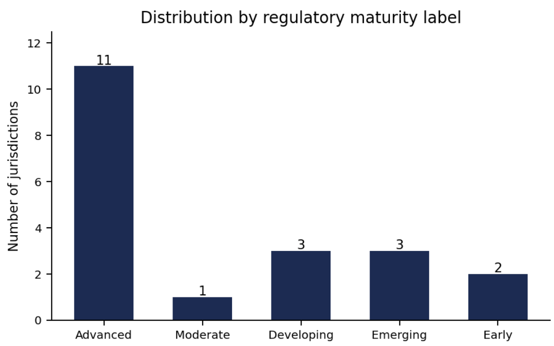

Figure 5.3: Distribution of jurisdictions by maturity label.

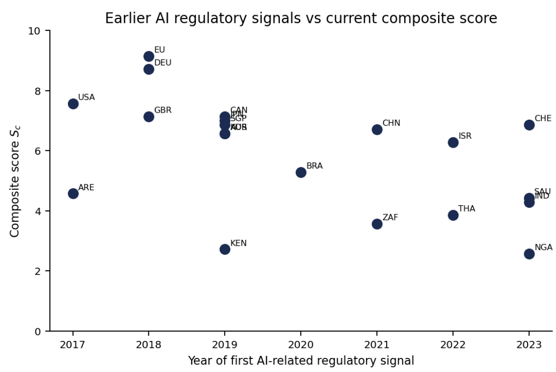

Figure 5.4: Year of first AI regulatory signal vs composite score.

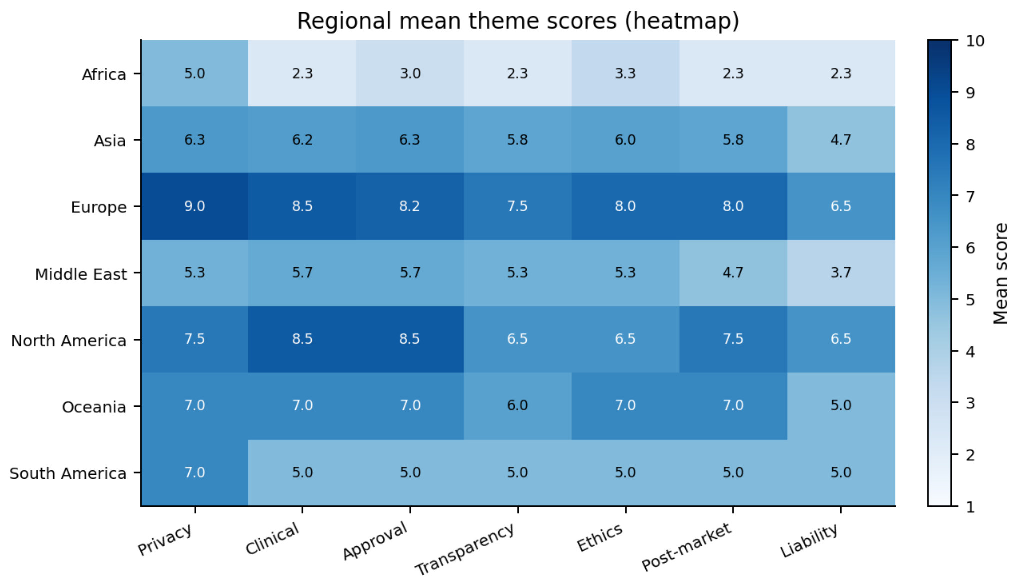

Figure 5.5: Regional mean theme scores visualized as a heatmap.

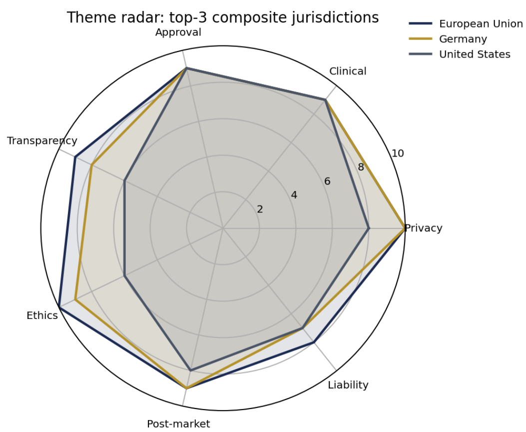

Figure 5.6: Radar comparison of the top-3 composite jurisdictions.

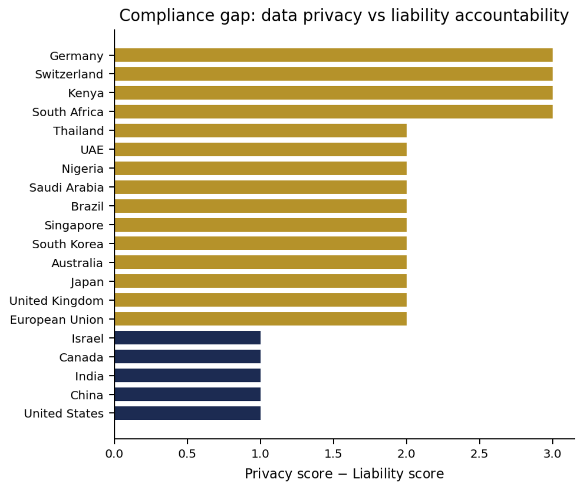

Figure 5.7: Privacy minus liability theme gap (gold: gap ≥ 2).

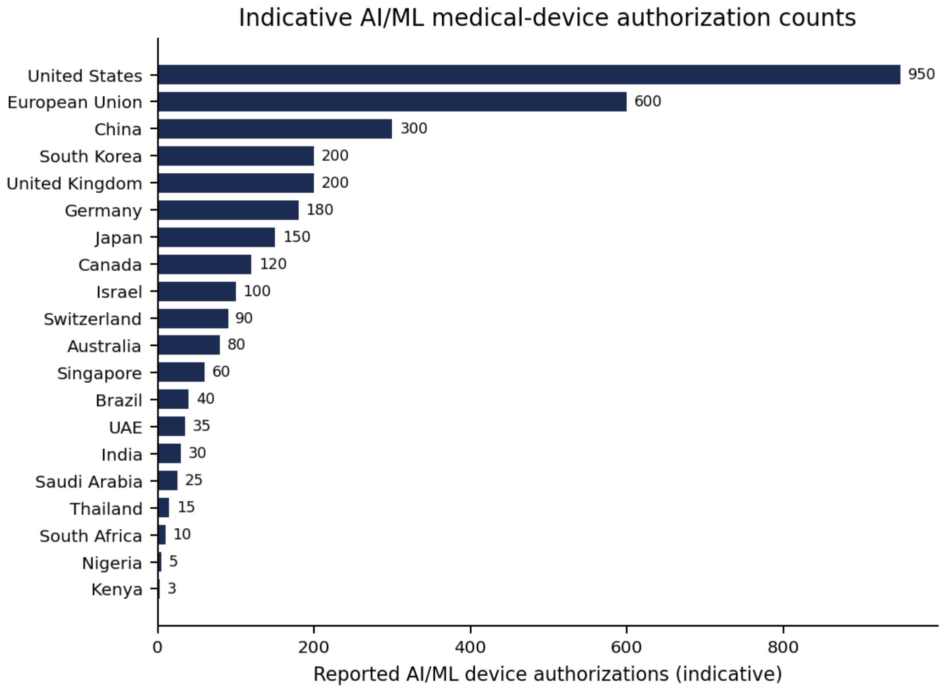

Figure 5.8: Indicative AI/ML medical-device authorization counts (not commensurate across agencies).

5.2 Interpretation of Global Patterns

Across the curated set, **data privacy** and **approval process** themes tend to score relatively high on average, reflecting the post-GDPR legislative wave and the near-universal presence of medical-device frameworks. **Liability** and, in several emerging jurisdictions, **post-market surveillance** and **transparency** lag—indicating that lifecycle governance and accountability remain frontier issues even where market-entry pathways exist. Figure 5.7 makes this asymmetry explicit: many jurisdictions score privacy substantially higher than liability.

Europe (including the EU aggregate and Germany) leads on composite scores, consistent with the layered GDPR + MDR + AI Act regime [14]. The United States leads on reported device throughput while showing comparatively moderate transparency/ethics scores, matching the literature's characterization of an innovation-forward, less horizontally mandated AI ethics posture [3], [4]. East Asian advanced jurisdictions combine substantial approval activity with evolving privacy and algorithm rules. African and some Middle Eastern entries illustrate data-protection progress preceding deep AI-SaMD capacity.

5.3 Regional Qualitative Analysis

5.3.1 North America

The USA–Canada pair demonstrates advanced agency science and GMLP leadership [19], with U.S. HIPAA sectoral privacy contrasting Canada's more comprehensive federal privacy trajectory. Fragmentation (U.S. state privacy; Canadian legislative timing) remains a practical compliance challenge for multi-province/multi-state deployments.

5.3.2 Europe

The EU AI Act is a watershed for binding high-risk obligations [14]. Germany's strong scores reflect both EU baselines and national digital-health instrumentation. The UK's post-Brexit pro-innovation stance creates dual-track complexity for vendors. Switzerland aligns closely with European data-protection expectations while maintaining distinct institutional pathways.

5.3.3 Asia–Pacific

China, Japan, South Korea, Singapore, Australia, India, and Thailand span advanced to developing maturity. Singapore's governance tooling and Japan's DASH framework are notable positive outliers relative to market size. India's large digital-health ambitions coexist with still-developing AI-device specificity. Data localization in China shapes multinational validation strategies [13].

5.3.4 Middle East and Africa

Israel combines innovation density with comparatively strong clinical-validation signals. Gulf states show rapid strategy publication with emerging operational detail. African jurisdictions score lower on AI-specific device governance but demonstrate meaningful privacy statutes and leapfrogging potential via mobile health—provided capacity-building accompanies legislation [24].

5.4 Country Profiles

The following subsections summarize curated narratives for each jurisdiction included in the dataset. They should be read together with theme scores and legislation lists in the JSON source of truth.

5.4.1 European Union

Region: Europe. **Maturity:** Advanced. **Primary regulator:** European Medicines Agency (EMA), National Competent Authorities, Notified Bodies. **Privacy law:** GDPR (General Data Protection Regulation). **AI-specific instrument:** EU AI Act (2024). **Device framework:** Medical Device Regulation (MDR 2017/745), In Vitro Diagnostic Regulation (IVDR 2017/746), CE Marking.

Approval and validation. Conformity assessment via Notified Bodies. Risk-based classification under MDR. EU AI Act classifies most healthcare AI as high-risk requiring conformity assessments, risk management, and human oversight. Clinical evaluation required under MDR. Post-market clinical follow-up (PMCF). EU AI Act mandates testing with real-world representative datasets, bias monitoring, and documentation.

Data governance and transparency. GDPR provides comprehensive data protection. Right to explanation for automated decisions (Article 22). Data Protection Impact Assessments (DPIA) required for high-risk processing. European Health Data Space (EHDS) proposed for secondary use of health data. EU AI Act mandates transparency obligations for high-risk AI: technical documentation, logging, human oversight. GDPR Article 22 provides right to meaningful information about automated decision logic.

Ethics, surveillance, and liability. EU AI Act embeds ethical principles: human oversight, non-discrimination, robustness. Ethics Guidelines for Trustworthy AI (HLEG, 2019). Fundamental Rights Impact Assessment required. Mandatory under MDR and EU AI Act. Post-market monitoring plans, serious incident reporting, periodic safety update reports. EU AI Liability Directive (proposed) creates presumption of causality for AI harm. Product Liability Directive revision to cover software/AI.

Theme scores (1–10). Privacy 10; Clinical validation 9; Approval 9; Transparency 9; Ethics 10; Post-market 9; Liability 8. Reported AI device approvals: 600.

Challenges and notable developments. Complex multi-layered regulatory landscape. Harmonization across 27 member states. Notified Body capacity constraints. Risk of over-regulation stifling innovation. EU AI Act is the world's first comprehensive AI law. High-risk healthcare AI must comply by 2026-2027. EHDS aims to create a unified health data framework.

5.4.2 Germany

Region: Europe. **Maturity:** Advanced. **Primary regulator:** BfArM (Federal Institute for Drugs and Medical Devices). **Privacy law:** GDPR + BDSG (Federal Data Protection Act). **AI-specific instrument:** EU AI Act (as EU member). National AI Strategy (2018, updated 2020).. **Device framework:** EU MDR implemented domestically. BfArM scientific advice for AI devices..

Approval and validation. EU MDR conformity assessment pathway. BfArM provides scientific advice. DiGA (Digital Health Applications) fast-track for prescribed digital therapies covered by statutory health insurance. DiGA requires evidence of positive healthcare effects. BfArM evaluation within 12 months. Real-world evidence accepted for DiGA provisional listing.

Data governance and transparency. GDPR + BDSG. Strict interpretation of data protection. Patient Data Protection Act (PDSG, 2021). Electronic health record (ePA) rollout. German Health Data Use Act (2024). EU AI Act transparency requirements. BDSG automated decision provisions. Strong culture of data protection enforcement.

Ethics, surveillance, and liability. Data Ethics Commission recommendations (2019). AI Observatory at BMAS. EU AI Act ethical framework. Strong academic ethics research tradition. EU MDR post-market surveillance. BfArM device vigilance. DiGA requires ongoing evidence generation. Product Liability Act. EU AI Liability Directive (when enacted). Medical malpractice law.

Theme scores (1–10). Privacy 10; Clinical validation 9; Approval 9; Transparency 8; Ethics 9; Post-market 9; Liability 7. Reported AI device approvals: 180.

Challenges and notable developments. Historically strict data protection culture can slow health data utilization. DiGA program growing pains. Interoperability of health IT systems. DiGA is the world's first regulatory pathway for prescribed digital therapies reimbursed by statutory health insurance. Health Data Use Act aims to unlock research data. Leading EU voice in AI regulation.

5.4.3 United States

Region: North America. **Maturity:** Advanced. **Primary regulator:** FDA (Food and Drug Administration). **Privacy law:** HIPAA (Health Insurance Portability and Accountability Act). **AI-specific instrument:** FDA AI/ML-Based SaMD Action Plan (2021). **Device framework:** FDA 510(k), De Novo, PMA pathways.

Approval and validation. Risk-based classification (Class I, II, III). AI/ML devices follow SaMD (Software as a Medical Device) framework. Predetermined Change Control Plan for adaptive algorithms. Clinical evidence required proportional to risk. Real-World Evidence (RWE) and Real-World Data (RWD) accepted. Good Machine Learning Practice (GMLP) principles published jointly with Health Canada and UK MHRA.

Data governance and transparency. HIPAA mandates protected health information (PHI) safeguards. De-identification standards under Safe Harbor and Expert Determination. HITECH Act strengthens enforcement. FDA encourages transparency and explainability but no binding mandate. Total Product Lifecycle (TPLC) approach for monitoring algorithm changes.

Ethics, surveillance, and liability. No federal AI ethics law. NIST AI Risk Management Framework (2023) provides voluntary guidance. Executive Order on AI (2023) directs agencies to manage AI risks. Mandatory adverse event reporting (MedWatch). Real-world performance monitoring encouraged under TPLC. Product liability under state tort law. FDA clearance does not shield from liability. Evolving case law on AI-related malpractice.

Theme scores (1–10). Privacy 8; Clinical validation 9; Approval 9; Transparency 6; Ethics 6; Post-market 8; Liability 7. Reported AI device approvals: 950.

Challenges and notable developments. Fragmented state-level privacy laws. No comprehensive federal AI legislation. Balancing innovation speed with safety oversight for adaptive algorithms. FDA has authorized over 950 AI/ML-enabled medical devices as of 2025. Predetermined Change Control Plan allows manufacturers to pre-specify algorithm modifications.

5.4.4 United Kingdom

Region: Europe. **Maturity:** Advanced. **Primary regulator:** MHRA (Medicines and Healthcare products Regulatory Agency). **Privacy law:** UK GDPR & Data Protection Act 2018. **AI-specific instrument:** Pro-innovation AI Regulation Framework (2023). **Device framework:** UK MDR 2002 (amended), UKCA marking.

Approval and validation. Risk-based device classification similar to EU MDR but diverging post-Brexit. MHRA Software and AI as Medical Device Change Programme. Proportionate regulation favoring innovation. NICE Evidence Standards Framework for digital health technologies. Multi-Agency Advisory Service (MAAS) provides coordinated guidance. GMLP principles co-developed with FDA and Health Canada.

Data governance and transparency. UK GDPR mirrors EU GDPR with domestic adjustments. NHS Data Strategy promotes secure data sharing. Caldicott Principles govern patient data confidentiality. Algorithmic Transparency Recording Standard for public sector. Pro-innovation framework uses existing regulators rather than new AI-specific rules.

Ethics, surveillance, and liability. Five cross-sector AI principles: safety, transparency, fairness, accountability, contestability. AI Safety Institute established (2023). No binding AI-specific ethics law. MHRA vigilance and post-market surveillance obligations. Yellow Card scheme for adverse incident reporting. Common law negligence and product liability. Government exploring AI-specific liability reforms.

Theme scores (1–10). Privacy 8; Clinical validation 8; Approval 7; Transparency 7; Ethics 7; Post-market 7; Liability 6. Reported AI device approvals: 200.

Challenges and notable developments. Post-Brexit regulatory divergence from EU. Balancing light-touch innovation-friendly approach with patient safety. Transition from CE to UKCA marking. UK positioned as pro-innovation alternative to EU AI Act. AI Safety Institute is a world-first government body for AI safety research. NHS AI Lab supports AI adoption in health services.

5.4.5 Canada

Region: North America. **Maturity:** Advanced. **Primary regulator:** Health Canada. **Privacy law:** PIPEDA (Personal Information Protection and Electronic Documents Act), Provincial health privacy laws (e.g., Ontario PHIPA). **AI-specific instrument:** AIDA (Artificial Intelligence and Data Act, proposed in Bill C-27). **Device framework:** Medical Devices Regulations (SOR/98-282). Risk-based classification (Class I-IV)..

Approval and validation. Risk-based classification. Pre-market review for Class II-IV. Health Canada guidance on SaMD and machine learning. GMLP principles co-developed with FDA and MHRA. Clinical evidence requirements proportional to risk. Acceptance of RWE. Health Canada actively engaged in international harmonization (IMDRF).

Data governance and transparency. PIPEDA governs private sector. Provincial laws (PHIPA, HIA) govern health information. Proposed Consumer Privacy Protection Act (CPPA) to modernize privacy framework. AIDA (proposed) would require transparency measures for high-impact AI. Directive on Automated Decision-Making for government systems (Algorithmic Impact Assessment).

Ethics, surveillance, and liability. Directive on Automated Decision-Making (2019) for federal government. AIDA (proposed) includes measures against biased AI. Pan-Canadian AI Strategy promotes responsible AI. Mandatory problem reporting. Post-market surveillance plans. Canada Vigilance Program. Common law negligence and product liability. AIDA (proposed) would introduce penalties for reckless/harmful AI deployment.

Theme scores (1–10). Privacy 7; Clinical validation 8; Approval 8; Transparency 7; Ethics 7; Post-market 7; Liability 6. Reported AI device approvals: 120.

Challenges and notable developments. Federal-provincial jurisdiction complexity. AIDA legislation delayed. Keeping pace with rapid AI advancement in healthcare. Canada co-developed GMLP with FDA and MHRA. Pan-Canadian Health Data Strategy in development. Strong AI research ecosystem (MILA, Vector, Amii) informing policy.

5.4.6 Japan

Region: Asia. **Maturity:** Advanced. **Primary regulator:** PMDA (Pharmaceuticals and Medical Devices Agency). **Privacy law:** APPI (Act on Protection of Personal Information, amended 2022). **AI-specific instrument:** AI Guidelines for Business (2024), Social Principles of Human-Centric AI (2019). **Device framework:** Pharmaceutical and Medical Device Act (PMD Act). PMDA DASH (Design, Appropriate use, Security, Health) for SaMD..

Approval and validation. Risk-based classification. IDATEN regulatory sandbox for innovative devices. PMDA Special Approval for Emergency and Priority Review pathways. Continuous learning AI framework under discussion. Clinical performance evaluation under PMD Act. PMDA guidance on AI medical device evaluation. Acceptance of foreign clinical data under ICH guidelines.

Data governance and transparency. APPI provides comprehensive data protection with special care for medical data. Next Generation Medical Infrastructure Act enables anonymized medical data use for research. Pseudonymized data processing provisions added in 2022 amendments. Social Principles of Human-Centric AI emphasize transparency and explainability. No binding legal mandate. AI Guidelines for Business (2024) provide sector-neutral guidance.

Ethics, surveillance, and liability. Human-centric AI principles adopted by government (2019). Council for Social Principles of Human-Centric AI. No binding AI ethics legislation. PMDA safety monitoring and re-examination. Good Vigilance Practice (GVP). Post-market clinical studies as needed. Product Liability Act covers defective products. AI-specific liability under discussion.

Theme scores (1–10). Privacy 7; Clinical validation 8; Approval 8; Transparency 6; Ethics 7; Post-market 8; Liability 5. Reported AI device approvals: 150.

Challenges and notable developments. Aging society drives demand but conservative regulatory culture. Language barriers in international harmonization. Slow adoption of digital health relative to technological capacity. Japan's regulatory sandbox (IDATEN) accelerates innovative device approval. Society 5.0 vision integrates AI in healthcare transformation. PMDA actively developing AI/ML device evaluation frameworks.

5.4.7 Singapore

Region: Asia. **Maturity:** Advanced. **Primary regulator:** HSA (Health Sciences Authority). **Privacy law:** PDPA (Personal Data Protection Act, 2012, amended 2021). **AI-specific instrument:** Model AI Governance Framework (2019, updated 2020), AI Verify (2022). **Device framework:** Health Products Act. Risk-based classification aligned with IMDRF..

Approval and validation. Risk-based classification. HSA regulatory sandbox initiatives. Aligned with IMDRF SaMD guidance. Streamlined approval for lower-risk devices. HSA clinical evidence requirements. Recognition of international approvals. Pragmatic approach balancing evidence rigor with access.

Data governance and transparency. PDPA governs data protection. Health Information Bill (proposed) for health-specific data governance. National Electronic Health Record (NEHR) framework. Model AI Governance Framework provides voluntary guidance on explainability. AI Verify is a testing framework for verifying AI governance claims.

Ethics, surveillance, and liability. Model AI Governance Framework emphasizes human-centric, explainable, fair AI. Advisory Council on Ethical Use of AI and Data. Smart Nation initiative integrates responsible AI. HSA post-market surveillance. Adverse event reporting. Periodic review of registered products. Common law product liability. Consumer Protection (Fair Trading) Act. No AI-specific liability provisions.

Theme scores (1–10). Privacy 7; Clinical validation 7; Approval 7; Transparency 8; Ethics 8; Post-market 6; Liability 5. Reported AI device approvals: 60.

Challenges and notable developments. Small domestic market limits local clinical evidence generation. Balancing business-friendly environment with robust oversight. Regional influence vs. global harmonization. AI Verify is the world's first AI governance testing framework. Singapore positioned as AI governance thought leader in ASEAN. Strong public-private partnerships in health AI.

5.4.8 Switzerland

Region: Europe. **Maturity:** Advanced. **Primary regulator:** Swissmedic. **Privacy law:** nFADP (New Federal Act on Data Protection, 2023). **AI-specific instrument:** No AI-specific law. Federal Council AI guidelines. Bilateral alignment with EU standards.. **Device framework:** Medical Devices Ordinance (MedDO) aligned with EU MDR. Mutual Recognition Agreement with EU..

Approval and validation. Aligned with EU conformity assessment. Swissmedic registration. Recognition of CE marking through bilateral agreements. Swiss market access closely tied to EU regulatory decisions. Clinical evidence requirements aligned with EU MDR. Strong clinical research infrastructure. International collaboration through IMDRF.

Data governance and transparency. nFADP (2023) modernized data protection aligned with GDPR adequacy. Health data as sensitive personal data. Cantonal health data regulations. Federal Council guidelines on AI transparency. No binding mandate. Swiss Digital Initiative Trust Label program.

Ethics, surveillance, and liability. Federal Ethics Committee on Non-Human Biotechnology. Swiss National Science Foundation AI ethics research. No binding AI ethics law.

Swissmedic vigilance. Aligned with EU post-market surveillance requirements. Adverse event reporting. Product Liability Act. Code of Obligations. No AI-specific liability provisions.

Theme scores (1–10). Privacy 8; Clinical validation 8; Approval 8; Transparency 6; Ethics 6; Post-market 7; Liability 5. Reported AI device approvals: 90.

Challenges and notable developments. Dependency on EU regulatory decisions without EU membership voting rights. Maintaining bilateral agreements post-EU AI Act. Small market size. Strong pharma/medtech hub (Novartis, Roche). nFADP brings Swiss privacy law to GDPR-equivalent. Active AI health research at ETH Zurich and EPFL.

5.4.9 China

Region: Asia. **Maturity:** Advanced. **Primary regulator:** NMPA (National Medical Products Administration). **Privacy law:** PIPL (Personal Information Protection Law, 2021). **AI-specific instrument:** Interim Measures for Generative AI (2023), Algorithm Recommendation Regulations (2022), Deep Synthesis Regulations (2023). **Device framework:** Regulations for Supervision and Administration of Medical Devices. Three-tier classification system. AI medical devices regulated as SaMD..

Approval and validation. Three-class risk-based system. NMPA has approved AI diagnostic devices in radiology, pathology, and ophthalmology. Expedited review pathways for innovative devices. Guiding Principles for AI Medical Device Registration (2021). Clinical trials required for Class II and III devices. Multi-center clinical validation emphasized. NMPA guidance on AI/ML algorithm validation and dataset requirements.

Data governance and transparency. PIPL governs personal information with strict consent and cross-border transfer rules. Data Security Law (2021) classifies health data as important data. Sensitive personal information protections for biometric and health data. Algorithm filing and registration required. Algorithmic Recommendation Regulations mandate transparency and user control. Deep synthesis (deepfake) regulations require labeling.

Ethics, surveillance, and liability. New Generation AI Ethics Code (2021). National AI governance principles emphasize harmony, fairness, and responsibility. State Council AI development plans integrate ethical guardrails. Mandatory adverse event reporting to NMPA. Periodic re-evaluation of registered devices. Quality management system audits. Civil Code provisions on product liability. Evolving jurisprudence on AI-caused harm. Algorithmic accountability provisions in algorithm regulations.

Theme scores (1–10). Privacy 7; Clinical validation 7; Approval 7; Transparency 7; Ethics 6; Post-market 7; Liability 6. Reported AI device approvals: 300.

Challenges and notable developments. Rapid regulatory evolution creates compliance complexity. Data localization requirements limit international collaboration. Balancing state surveillance priorities with individual privacy. China is among the first to regulate algorithmic recommendations and generative AI. NMPA rapidly approving AI diagnostic tools. National AI standardization efforts underway.

5.4.10 Australia

Region: Oceania. **Maturity:** Moderate. **Primary regulator:** TGA (Therapeutic Goods Administration). **Privacy law:** Privacy Act 1988 (amended), Australian Privacy Principles (APPs). **AI-specific instrument:** Australia's AI Ethics Principles (2019), Voluntary AI Safety Standard (2024). **Device framework:** Therapeutic Goods Act 1989. Risk-based classification aligned with IMDRF. TGA regulatory guidance on SaMD..

Approval and validation. Risk-based classification. TGA registration and conformity assessment. Alignment with international standards. SaMD-specific regulatory guidance. Clinical evidence requirements. TGA guidance on digital health technologies. Recognition of international clinical evidence.

Data governance and transparency. Privacy Act with Australian Privacy Principles. My Health Records Act for digital health records. Health data subject to enhanced protections. Voluntary AI Ethics Principles include transparency. No binding mandate. Government response to AI consultation ongoing.

Ethics, surveillance, and liability. Eight AI Ethics Principles (2019): human-centric, fair, transparent, contestable, accountable, reliable, safe, privacy-protected. Voluntary adoption. National AI Centre provides guidance. TGA post-market review and vigilance. Adverse event reporting mandatory. Systematic review of registered products. Australian Consumer Law product liability. Professional negligence standards. No AI-specific liability law.

Theme scores (1–10). Privacy 7; Clinical validation 7; Approval 7; Transparency 6; Ethics 7; Post-market 7; Liability 5. Reported AI device approvals: 80.

Challenges and notable developments. Voluntary nature of AI ethics framework limits enforcement. Resource constraints in TGA for AI evaluation. Geographic barriers to clinical trial recruitment. Australia's AI Ethics Principles among the earliest globally. TGA actively developing SaMD regulatory pathways. National Digital Health Strategy driving AI integration.

5.4.11 South Korea

Region: Asia. **Maturity:** Advanced. **Primary regulator:** MFDS (Ministry of Food and Drug Safety). **Privacy law:** PIPA (Personal Information Protection Act, amended 2023). **AI-specific instrument:** AI Framework Act (proposed), National AI Strategy (2019). **Device framework:** Medical Devices Act. Risk-based classification. Regulatory Science Center for AI/ML devices..

Approval and validation. Risk-based classification. MFDS has fast-track pathways for innovative AI devices. Regulatory Science Center specializing in AI medical device review. Software-specific classification criteria. Clinical trial requirements under Medical Devices Act. MFDS guidance on AI/ML medical device clinical validation. Growing emphasis on real-world evidence.

Data governance and transparency. PIPA provides comprehensive data protection. 2023 amendments enable pseudonymized data processing for research. MyData framework for personal data portability. National AI Strategy emphasizes trustworthy AI. No binding transparency mandate yet. AI Framework Act (proposed) would establish transparency requirements.

Ethics, surveillance, and liability. National AI Ethics Standards (2020). AI Ethics Guidelines emphasize human dignity, public good, and technical soundness. No binding legislation yet. MFDS post-market surveillance. Adverse event reporting system. Quality management requirements. Product Liability Act. Civil liability framework. AI-specific liability considerations under development.

Theme scores (1–10). Privacy 7; Clinical validation 7; Approval 8; Transparency 6; Ethics 6; Post-market 7; Liability 5. Reported AI device approvals: 200.

Challenges and notable developments. Balancing rapid technological adoption with regulatory oversight. Harmonizing domestic standards with international frameworks. SME compliance burden. South Korea has one of the highest AI medical device approval rates globally. MFDS Regulatory Science Center is a dedicated AI device evaluation unit. K-BioHealth initiative driving AI healthcare innovation.

5.4.12 Israel

Region: Middle East. **Maturity:** Advanced. **Primary regulator:** MOH AMAR (Division of Medical Devices, Ministry of Health). **Privacy law:** Privacy Protection Law (1981, amended), Health Information Exchange regulations. **AI-specific instrument:** AI Policy and Ethics Framework (2022), National AI Program. **Device framework:** Medical Device Regulations under Pharmacists' Ordinance. Alignment with EU MDR and FDA pathways..

Approval and validation. Recognized market approval from FDA, CE, or other reference agencies. AMAR reviews and registers. Expedited pathways for innovative technologies. Growing AI device registrations. Clinical trial requirements under MOH. Strong real-world data infrastructure. International clinical validation partnerships.

Data governance and transparency. Privacy Protection Law with sector-specific health regulations. Unique national health data infrastructure (Clalit, Maccabi, etc.). Rich longitudinal health datasets enabling AI research. AI Policy Framework recommends transparency. No binding mandate. Active academic and industry discourse on AI governance.

Ethics, surveillance, and liability. AI Ethics Framework addresses healthcare applications. Innovation Authority promotes responsible AI. Active ethics research community. MOH post-market surveillance. Adverse event reporting. Device vigilance system. Tort liability. Patient Rights Law. No AI-specific liability provisions. Active legal scholarship on the topic.

Theme scores (1–10). Privacy 6; Clinical validation 8; Approval 7; Transparency 6; Ethics 6; Post-market 6; Liability 5. Reported AI device approvals: 100.

Challenges and notable developments. Outdated base privacy legislation. Balancing startup-friendly ecosystem with regulatory rigor. Small domestic market necessitates international focus. Israel is a global health AI innovation hub. Unique national health data infrastructure enables world-class AI research. Highest health AI startups per capita globally.

5.4.13 Brazil

Region: South America. **Maturity:** Developing. **Primary regulator:** ANVISA (Agência Nacional de Vigilância Sanitária). **Privacy law:** LGPD (Lei Geral de Proteção de Dados, 2020). **AI-specific instrument:** AI Bill (PL 2338/2023, under legislative process). **Device framework:** RDC 185/2001 and updates. Risk-based classification (Classes I-IV)..

Approval and validation. Risk-based classification. ANVISA registration required. SaMD guidance under development. International harmonization through IMDRF participation. Clinical evidence requirements under ANVISA regulations. Limited AI-specific validation guidance. Growing clinical AI research community.

Data governance and transparency. LGPD provides comprehensive data protection modeled on GDPR. Health data classified as sensitive. ANPD (National Data Protection Authority) oversees enforcement. AI Bill (proposed) includes transparency obligations for high-risk AI. LGPD provides right to review of automated decisions. No current binding AI transparency mandate.

Ethics, surveillance, and liability. AI Bill (proposed) incorporates ethical principles: human oversight, non-discrimination, transparency. Brazilian AI Strategy (2021) promotes responsible development. ANVISA post-market surveillance. Tecnovigilância (technovigilance) system. Mandatory adverse event reporting. Consumer Defense Code. Civil liability framework. AI Bill (proposed) would establish risk-based liability for AI.

Theme scores (1–10). Privacy 7; Clinical validation 5; Approval 5; Transparency 5; Ethics 5; Post-market 5; Liability 5. Reported AI device approvals: 40.

Challenges and notable developments. AI regulatory framework still in legislative process. Limited ANVISA capacity for AI device evaluation. Digital health infrastructure gaps in underserved regions. LGPD is Latin America's most comprehensive data protection law. Active AI legislation process. Growing health AI startup ecosystem in São Paulo.

5.4.14 UAE

Region: Middle East. **Maturity:** Emerging. **Primary regulator:** MOH (Ministry of Health and Prevention), HAAD, DHA. **Privacy law:** Federal Decree-Law No. 45/2021 (Personal Data Protection). **AI-specific instrument:** National AI Strategy 2031, Minister of State for AI appointed (2017). **Device framework:** MOH medical device registration. Emirates Conformity Assessment Scheme (ECAS)..

Approval and validation. MOH registration with recognition of international approvals (FDA, CE). Emirates Authority for Standardization (ESMA) standards. Dubai Health Authority (DHA) parallel pathway. Expedited processes for innovative technologies. Clinical evidence requirements for device registration. Recognition of international clinical data. Mohamed bin Rashid University of Medicine and Health Sciences research programs.

Data governance and transparency. Federal data protection law (2021). Dubai International Financial Centre (DIFC) Data Protection Law. Abu Dhabi Healthcare Information and Cyber Security Standard. Dubai Health Data Policy. National AI Strategy includes governance principles. No binding transparency mandate. AI governance guidelines under development.

Ethics, surveillance, and liability. National AI Strategy ethical guidelines. UAE AI Ethics Board. Oxford-UAE AI research partnership on responsible AI. MOH device vigilance. Adverse event reporting. Quality management requirements. Civil Code liability provisions. Consumer Protection Law. No AI-specific liability framework.

Theme scores (1–10). Privacy 5; Clinical validation 5; Approval 5; Transparency 5; Ethics 5; Post-market 4; Liability 3. Reported AI device approvals: 35.

Challenges and notable developments. Multi-authority regulatory landscape (federal and emirate-level). Rapid strategy ambitions vs. regulatory maturity. Building domestic healthcare data infrastructure. First country to appoint a Minister of State for AI (2017). Ambitious National AI Strategy 2031. Hub for health AI startups and international collaborations.

5.4.15 Saudi Arabia

Region: Middle East. **Maturity:** Emerging. **Primary regulator:** SFDA (Saudi Food and Drug Authority). **Privacy law:** PDPL (Personal Data Protection Law, 2023). **AI-specific instrument:** SDAIA AI Ethics Principles (2023), National AI Strategy. **Device framework:** Medical Devices Interim Regulation. SFDA SaMD guidance. Alignment with IMDRF..

Approval and validation. SFDA registration and conformity assessment. Recognition of international approvals (FDA, CE). Expedited review pathways for innovative products. SaMD regulatory framework under development. Clinical evidence requirements. Recognition of international clinical data. National Center for AI (NCAI) supports validation research.

Data governance and transparency. PDPL (effective 2023) establishes comprehensive data protection. Health data classified as sensitive. Cloud Computing Regulatory Framework governs data hosting. SDAIA AI Ethics Principles include transparency. No binding mandate. Data and AI governance frameworks under development.

Ethics, surveillance, and liability. SDAIA AI Ethics Principles (2023): fairness, transparency, accountability, privacy, security, human-centricity. Vision 2030 AI integration goals. SFDA post-market surveillance requirements. Adverse event reporting. Medical device vigilance system. Civil liability framework. AI-specific liability not yet addressed. Consumer protection provisions apply.

Theme scores (1–10). Privacy 5; Clinical validation 4; Approval 5; Transparency 5; Ethics 5; Post-market 4; Liability 3. Reported AI device approvals: 25.

Challenges and notable developments. Regulatory framework still maturing. Building domestic regulatory expertise. Dependency on international recognition pathways. Limited public health data infrastructure. Massive AI investment under Vision 2030. SDAIA established as dedicated AI authority. NEOM project integrating AI healthcare. Rapid digital health transformation.

5.4.16 India

Region: Asia. **Maturity:** Developing. **Primary regulator:** CDSCO (Central Drugs Standard Control Organisation). **Privacy law:** Digital Personal Data Protection Act (DPDP, 2023). **AI-specific instrument:** No AI-specific law. NITI Aayog AI Strategy and Responsible AI guidelines (2021).. **Device framework:** Medical Devices Rules 2017 (amended 2020). SaMD classification under development..

Approval and validation. Risk-based classification (Class A-D). CDSCO registration required. Limited specific guidance for AI/ML medical devices. Reliance on international standards (ISO, IEC). Clinical investigation requirements under Medical Devices Rules. Limited AI-specific validation guidance. Growing clinical AI research ecosystem.

Data governance and transparency. DPDP Act 2023 establishes consent-based framework with data fiduciary obligations. Health data treated as sensitive. Ayushman Bharat Digital Mission (ABDM) creates digital health infrastructure. No binding transparency requirements. NITI Aayog principles recommend explainability and accountability. IndiaAI Mission (2024) promotes responsible AI.

Ethics, surveillance, and liability. NITI Aayog Responsible AI principles (fairness, accountability, transparency, privacy, safety). No binding ethics law for AI. ICMR National Ethical Guidelines for Biomedical and Health Research. Post-market surveillance requirements under Medical Devices Rules. Materiovigilance Programme of India (MvPI) for adverse event reporting. Consumer Protection Act 2019 covers product defects. IT Act provisions for data breaches. No AI-specific liability framework.

Theme scores (1–10). Privacy 5; Clinical validation 4; Approval 4; Transparency 4; Ethics 5; Post-market 4; Liability 4. Reported AI device approvals: 30.

Challenges and notable developments. Regulatory framework still maturing for AI devices. Limited CDSCO capacity for AI/ML evaluation. Infrastructure gaps in digital health. Balancing innovation needs with regulatory rigor. DPDP Act 2023 is India's first comprehensive data protection law. Ayushman Bharat Digital Mission building foundational digital health infrastructure. IndiaAI Mission allocating significant resources for AI development.

5.4.17 Thailand

Region: Asia. **Maturity:** Developing. **Primary regulator:** Thai FDA (Food and Drug Administration). **Privacy law:** PDPA (Personal Data Protection Act, 2022). **AI-specific instrument:** National AI Strategy and Action Plan (2022-2027). No binding AI law..

Device framework: Medical Device Act B.E. 2551 (2008). Risk-based classification. SaMD guidance under development..

Approval and validation. Thai FDA registration. Risk-based classification. Limited AI-specific device guidance. ASEAN Medical Device Directive harmonization. Clinical evidence requirements. Thai Clinical Trials Registry. Limited AI-specific validation frameworks. Growing research capacity.

Data governance and transparency. PDPA provides comprehensive data protection effective 2022. Health data as sensitive data. Ministry of Public Health data governance policies. National AI Ethics Guidelines include transparency. No binding mandate. NECTEC AI research addresses explainability.

Ethics, surveillance, and liability. National AI Ethics Guidelines. Thailand 4.0 policy includes responsible technology principles. No binding AI ethics law. Thai FDA post-market surveillance. Adverse event reporting. Resource-constrained monitoring. Consumer Protection Act. Product Liability Act. No AI-specific liability.

Theme scores (1–10). Privacy 5; Clinical validation 4; Approval 4; Transparency 4; Ethics 4; Post-market 3; Liability 3. Reported AI device approvals: 15.

Challenges and notable developments. Regulatory framework for AI devices still developing. Digital health infrastructure building. Balancing medical tourism innovation with domestic healthcare priorities. PDPA implementation is a major privacy milestone. Thailand 4.0 drives digital health investment. Medical tourism sector catalyzing AI adoption.

5.4.18 South Africa

Region: Africa. **Maturity:** Emerging. **Primary regulator:** SAHPRA (South African Health Products Regulatory Authority). **Privacy law:** POPIA (Protection of Personal Information Act, 2021). **AI-specific instrument:** No AI-specific regulation. AI Policy Framework under development.. **Device framework:** Medicines and Related Substances Act. Medical device regulations under SAHPRA. Classification system under implementation..

Approval and validation. SAHPRA registration transitioning from previous MCC system. Medical device regulatory framework being modernized. Limited AI-specific device guidance. Backlog in device registrations. Clinical evidence requirements under development. Reliance on international clinical data. Limited local AI validation infrastructure.

Data governance and transparency. POPIA provides comprehensive data protection. Health data classified as special personal information. Conditions for lawful processing include consent and public health exceptions. No binding transparency requirements. Presidential Commission on 4IR (2020) recommended AI governance. Draft AI Policy Framework under consultation.

Ethics, surveillance, and liability. Presidential Commission on 4IR recommendations. No binding AI ethics framework. Research ethics committees govern clinical AI research. Ubuntu philosophy informing ethical discourse. SAHPRA post-market surveillance developing. Adverse event reporting being strengthened. Resource limitations affect monitoring capacity. Consumer Protection Act. Common law product liability. No AI-specific liability provisions.

Theme scores (1–10). Privacy 6; Clinical validation 3; Approval 3; Transparency 3; Ethics 4; Post-market 3; Liability 3. Reported AI device approvals: 10.

Challenges and notable developments. SAHPRA capacity and resource constraints. Digital infrastructure gaps. Brain drain in health AI expertise. Balancing innovation with addressing basic healthcare access. POPIA is Africa's most comprehensive data protection law. CSIR and universities driving AI health research. Growing health tech ecosystem in Cape Town and Johannesburg.

5.4.19 Kenya

Region: Africa. **Maturity:** Early. **Primary regulator:** PPB (Pharmacy and Poisons Board). **Privacy law:** Data Protection Act (2019). **AI-specific instrument:** No AI-specific regulation. Blockchain and AI Taskforce Report (2019).. **Device framework:** PPB medical device regulations. Classification aligned with WHO guidelines. Limited SaMD framework..

Approval and validation. PPB registration for medical devices. Limited AI-specific guidance. Recognition of WHO prequalified products. Growing regulatory capacity through international partnerships. National Commission for Science, Technology and Innovation (NACOSTI) oversees research. Clinical trial regulations. Limited AI-specific validation frameworks.

Data governance and transparency. Data Protection Act 2019 with Office of Data Protection Commissioner (ODPC). Health data as sensitive data category. Kenya Health Policy framework guides health data governance. No binding transparency requirements. Blockchain and AI Taskforce recommendations (2019) address governance. Limited implementation.

Ethics, surveillance, and liability. Blockchain and AI Taskforce ethical recommendations. No binding AI ethics framework. KEMRI Ethics Review Committee for health research. PPB pharmacovigilance. Limited medical device post-market monitoring. Resource constraints. Consumer Protection Act. Tort law framework. No AI-specific liability provisions.

Theme scores (1–10). Privacy 5; Clinical validation 2; Approval 3; Transparency 2; Ethics 3; Post-market 2; Liability 2. Reported AI device approvals: 3.

Challenges and notable developments. Limited regulatory capacity. Infrastructure constraints. Prioritizing basic healthcare access alongside innovation. Dependency on international standards and partnerships. Kenya is a mobile health innovation leader in Africa. Data Protection Act among earliest in East Africa. Growing AI-for-health research at universities and tech hubs.

5.4.20 Nigeria

Region: Africa. **Maturity:** Early. **Primary regulator:** NAFDAC (National Agency for Food and Drug Administration and Control). **Privacy law:** NDPR (Nigeria Data Protection Regulation, 2019) / NDPA (Nigeria Data Protection Act, 2023). **AI-specific instrument:** National AI Strategy (draft). No binding AI regulation.. **Device framework:** NAFDAC medical device registration. Classification system. Limited SaMD-specific framework..

Approval and validation. NAFDAC registration for medical devices. Limited specific guidance for AI/ML devices. Reliance on international approvals and standards. Limited AI-specific clinical validation requirements. Research ethics committees govern clinical studies. Growing AI-for-health research community.

Data governance and transparency. NDPA 2023 establishes comprehensive data protection. Health data as sensitive category. NITDA (National Information Technology Development Agency) oversees digital policy. No binding transparency requirements. National Digital Economy Policy addresses some digital governance aspects.

Ethics, surveillance, and liability. No binding AI ethics framework. Academic and civil society discourse on AI ethics. NITDA strategic framework includes responsible AI elements. NAFDAC pharmacovigilance. Limited medical device post-market monitoring. Resource constraints affect surveillance capacity. Consumer protection laws. Federal Competition and Consumer Protection Act. No AI-specific liability.

Theme scores (1–10). Privacy 4; Clinical validation 2; Approval 3; Transparency 2; Ethics 3; Post-market 2; Liability 2. Reported AI device approvals: 5.

Challenges and notable developments. Limited regulatory capacity for AI devices. Digital infrastructure gaps. Healthcare access challenges precede AI adoption. Funding constraints for regulatory modernization. NDPA 2023 is a significant step in data governance. Growing AI research community (e.g., Data Science Nigeria). Mobile health innovations bypassing traditional regulatory pathways.

5.5 Global Trends Linked to Clinical Use Cases

Table 5.7: Global trends in AI healthcare regulation

Trend	Description	Adoption	Since
Convergence on Risk-Based Regulation	Most jurisdictions adopting risk-based classification of AI medical devices, influenced by IMDRF framework.	High	2017
Comprehensive Data Protection Laws	Post-GDPR wave of national data protection laws with health data as sensitive category.	High	2018
AI-Specific Legislation	Growing trend toward dedicated AI laws (EU AI Act leading, followed by Brazil, Canada, South Korea proposals).	Medium	2021
International Harmonization Efforts	IMDRF, GMLP (FDA-Health Canada-MHRA), and WHO guidance driving convergence.	Medium	2019
Ethical AI Frameworks	Proliferation of national AI ethics principles, mostly voluntary, increasingly being legislated.	High	2018
Regulatory Sandboxes for Health AI	Controlled testing environments for innovative AI devices before full regulatory compliance.	Low	2020
Real-World Evidence Acceptance	Growing regulatory acceptance of real-world data and evidence for AI device evaluation.	Medium	2019
Adaptive Algorithm Governance	Frameworks for managing continuously learning AI systems (e.g., FDA Predetermined Change Control Plan).	Low	2021
AI Adoption in Radiology and Digital Pathology Workflows	AI image analysis is increasingly deployed for triage and decision support in radiology and pathology, driving demand for clinical validation, transparency for clinicians, and post-deployment monitoring.	High	2021
Genomic AI for Precision Medicine (Data Governance Focus)	Genomic interpretation and clinical decision support are expanding, with governance emphasizing privacy, representative datasets, and controlled cross-border data use.	Medium	2020

Trend	**Description**	**Adoption**	**Since**
AI-Augmented Drug Discovery and Clinical Development Support	AI supports discovery and development workflows and increasingly influences clinical evidence generation, increasing focus on documentation, auditability, and liability assumptions.	Medium	2021
Hospital Deployment of Clinical Decision Support (Workflow Integration)	Hospitals and clinics integrate AI into electronic workflows, making Total Product Lifecycle governance and user training central to safe deployment.	High	2022
Public Health Surveillance and Outbreak Support with AI	AI systems are being explored for population-level surveillance and outbreak support, requiring accountability mechanisms because impacts extend beyond individual patients.	Medium	2020

- **Convergence on Risk-Based Regulation.** Most jurisdictions adopting risk-based classification of AI medical devices, influenced by IMDRF framework. (adoption: High; emerged: 2017).
- **Comprehensive Data Protection Laws.** Post-GDPR wave of national data protection laws with health data as sensitive category. (adoption: High; emerged: 2018).
- **AI-Specific Legislation.** Growing trend toward dedicated AI laws (EU AI Act leading, followed by Brazil, Canada, South Korea proposals). (adoption: Medium; emerged: 2021).
- **International Harmonization Efforts.** IMDRF, GMLP (FDA-Health Canada-MHRA), and WHO guidance driving convergence. (adoption: Medium; emerged: 2019).
- **Ethical AI Frameworks.** Proliferation of national AI ethics principles, mostly voluntary, increasingly being legislated. (adoption: High; emerged: 2018).
- **Regulatory Sandboxes for Health AI.** Controlled testing environments for innovative AI devices before full regulatory compliance. (adoption: Low; emerged: 2020).
- **Real-World Evidence Acceptance.** Growing regulatory acceptance of real-world data and evidence for AI device evaluation. (adoption: Medium; emerged: 2019).
- **Adaptive Algorithm Governance.** Frameworks for managing continuously learning AI systems (e.g., FDA Predetermined Change Control Plan). (adoption: Low; emerged: 2021).
- **AI Adoption in Radiology and Digital Pathology Workflows.** AI image analysis is increasingly deployed for triage and decision support in radiology and pathology, driving demand for clinical validation, transparency for clinicians, and post-deployment monitoring. (adoption: High; emerged: 2021).

- **Genomic AI for Precision Medicine (Data Governance Focus).** Genomic interpretation and clinical decision support are expanding, with governance emphasizing privacy, representative datasets, and controlled cross-border data use. (adoption: Medium; emerged: 2020).
- **AI-Augmented Drug Discovery and Clinical Development Support.** AI supports discovery and development workflows and increasingly influences clinical evidence generation, increasing focus on documentation, auditability, and liability assumptions. (adoption: Medium; emerged: 2021).
- **Hospital Deployment of Clinical Decision Support (Workflow Integration).** Hospitals and clinics integrate AI into electronic workflows, making Total Product Lifecycle governance and user training central to safe deployment. (adoption: High; emerged: 2022).
- **Public Health Surveillance and Outbreak Support with AI.** AI systems are being explored for population-level surveillance and outbreak support, requiring accountability mechanisms because impacts extend beyond individual patients. (adoption: Medium; emerged: 2020).

These trends justify the dashboard's use-case readiness views: regulatory maturity is not abstract—it constrains radiology triage tools, genomic pipelines, hospital CDS, and public-health analytics differently [2], [17].

5.6 Dashboard Outcome Assessment

The implemented Streamlit artifact is published at `https://github.com/remi7025/AI-Healthcare-Compliance-Dashboard` and loads `data/compliance_dataset.json` at runtime. Relative to Phase 1 goals, the system provides multi-level presentation (global map $\rightarrow$ theme heatmaps $\rightarrow$ country narratives), comparative radar/bar views, and in-app literature browsing. Real-time monitoring of live legislative feeds remains roadmap work; the current release emphasizes curated correctness over continuous scraping. HITL remains institutionalized via manual score changes under version control.

Table 5.8: Dashboard tabs implemented in the open-source Streamlit application

Tab	Function
World Map	Choropleth of regulatory maturity and selected theme scores
Country Comparison	Side-by-side radar charts, bar charts, and score tables
Theme Analysis	Heatmap, regional averages, and privacy–liability gap views
Global Trends	Timeline cards and device-authorization distribution
Country Details	Deep-dive profiles with expandable regulatory sections
Literature Review	Full thematic review rendered inside the dashboard

Interactive filters support region, maturity level, and theme focus; tables and charts can be exported for audit. The figures in Section 5.1 are static counterparts of these interactive views, regenerated from the same JSON source of truth so that the written report and the repository remain aligned.

5.7 Discussion and Critical Analysis

5.7.1 Strengths

The seven-theme ontology surfaces trade-offs invisible in single rankings; open JSON enables reuse; dual UI implementations support both rapid prototyping and presentation polish; and the bibliography combines peer-reviewed studies with primary regulatory sources.

5.7.2 Limitations and Threats to Validity

- Expert scores entail subjectivity despite calibration rules;
- English-language source bias may under-represent local-language guidance;
- Device counts are imperfectly comparable across agencies;
- EU-as-aggregate smooths member-state variance;
- NLP classification is designed more than fully trained in this draft;
- Scores are snapshots; law changes can invalidate cells without longitudinal versioning discipline.

5.7.3 Implications

For developers, the results argue for modular evidence packages mapped to theme gaps in target markets. For policymakers, capacity-building in post-market and liability themes may yield safer scaling than approval acceleration alone. For academia, open comparative datasets can ground future supervised NLP benchmarks—addressing the "gold standard" gap noted in Phase 1.

5.8 Sensitivity and Robustness Checks

To probe whether rankings are brittle to theme weighting, three alternative composites were inspected conceptually (and can be recomputed from the exported CSV):

1. **Privacy-heavy:** double weight on data privacy and transparency;
2. **Safety-heavy:** double weight on clinical validation and post-market surveillance;
3. **Market-access-heavy:** double weight on approval process.

Qualitatively, European jurisdictions remain near the top under privacy-heavy and safety-heavy schemes, while high-throughput markets can rise under market-access-heavy schemes without matching transparency scores. This confirms the design choice to expose theme-level detail rather than a single opaque index.

5.9 Implications for NLP-Assisted Compliance Monitoring

The results also inform the NLP roadmap. High-variance themes (liability, transparency in emerging markets) are exactly where clause-level retrieval would help curators most: locating sparse signals in long strategy documents. Conversely, privacy scores are often anchored by a small number of well-known statutes, so NLP gains are smaller. A practical annotation strategy is therefore theme-stratified: oversample liability and post-market clauses when building the first labeled corpus.

5.10 Stakeholder Demo Scenarios

Three demo scenarios illustrate how the dashboard supports decision-oriented exploration:

1. compare EU vs USA on transparency and ethics;
2. filter African jurisdictions and discuss privacy-first trajectories;
3. open a radiology use-case readiness view and relate it to validation/post-market themes.

These scenarios connect quantitative tables to operational storytelling for policymakers, developers, and hospital stakeholders.

5.11 Extended Comparative Notes on Enforcement Capacity

A recurring finding across the curated narratives is that *law density* and *enforcement capacity* are imperfectly correlated. Jurisdictions may publish ambitious AI strategies while lacking specialized reviewers for adaptive SaMD, or conversely may run mature device agencies with limited horizontal AI statutes. The seven-theme scorecard attempts to capture both dimensions through narrative fields and maturity labels, but quantitative capacity metrics (reviewer headcount, average review time, inspection frequency) remain sparsely published. Future dataset versions should prioritize such operational indicators wherever agencies disclose them.

5.12 Reading Device Approval Counts with Caution

Reported AI/ML device authorization counts are useful signals of market activity and agency throughput, yet they are not commensurate across borders. Definitions of "AI-enabled," inclusion of radiology vs non-radiology products, and public list completeness vary. In the results tables, counts therefore appear beside theme scores rather than inside the composite S_c. A high count with moderate transparency scores suggests a different risk posture than a moderate count with strong post-market obligations. Decision-makers should avoid using counts as a sole proxy for "regulatory quality."

5.13 Toward Longitudinal Governance Analytics

The present release is a cross-sectional snapshot. Legislative change events—entry into force of the EU AI Act high-risk duties, new FDA guidance documents, national AI bills—should eventually appear as time-stamped deltas on each theme. Even a simple changelog (date, country, theme, old score, new score, justification) would transform the dashboard from a static map into a monitoring instrument aligned with Phase 1's real-time ambition, while preserving HITL control over every numeric change.

5.14 Detailed UI Task Walkthroughs

5.14.1 Task A — Identify a transparency gap

A user filters to Advanced maturity jurisdictions, opens Theme Analysis, and sorts by transparency. Comparing the United States and the European Union illustrates how high device throughput can coexist with comparatively moderate mandated transparency, whereas EU high-risk duties elevate documentation and logging expectations [4], [14].

5.14.2 Task B — Explore an emerging-market trajectory

A user filters to Africa, opens Country Details for South Africa, Nigeria, and Kenya, and notes stronger relative privacy statutes than AI-specific SaMD pathways. The narrative fields highlight leapfrogging opportunities via mobile health alongside capacity constraints [24].

5.14.3 Task C — Connect regulation to radiology deployment

A user opens AI Use Cases, selects radiology readiness, and inspects which themes drive the derived score. Clinical validation and post-market weights dominate, matching literature concerns about external validation and monitoring after PACS integration [3], [17].

5.15 Export and Audit Trail

CSV export of the filtered country table enables external audit in spreadsheets or statistical software. Recommended audit columns include country, region, maturity, each theme score, composite, and device count. Analysts should archive the dataset `version` and `last_updated` fields with any exported figure to preserve provenance.

5.16 Limitations Specific to Visualization

Choropleths can exaggerate geographic area relative to regulatory importance; radar charts can imply equal theme spacing that is conceptually ordinal rather than interval. The report therefore privileges tables for precise values and uses charts for pattern discovery. Color palettes should remain color-blind friendly in future visual refinements.

5.17 Ethical Communication of Scores

When presenting results, a composite ranking view should always be paired with a theme breakdown, together with an explicit non-legal-advice caveat. Comparative governance research fails ethically if numbers travel without their definitions.

6. Conclusion and Future Work

6.1 Summary of Contributions

This Final Year Project asked how multi-jurisdictional AI healthcare regulatory requirements can be systematically synthesized, scored, and interactively compared. The work answers that question through three integrated contributions: (i) a thematic state-of-the-art connecting SaMD, privacy, validation, transparency, ethics, surveillance, and liability; (ii) a curated twenty-jurisdiction dataset with an explicit seven-theme scoring ontology; and (iii) an interactive dashboard—open-sourced at `https://github.com/remi7025/AI-Healthcare-Compliance-Dashboard`—that turns those indicators into explorable maps, comparisons, trends, and clinical use-case views.

Relative to the SMART objectives in Chapter 1, the project delivered measurable theme scores and device-count indicators, an achievable local dashboard without proprietary APIs, and a time-bound full report aligned with the aivancity PFE structure. Scientifically, the contribution is an open comparative ontology plus interactive analytics; operationally, the contribution is a briefing-ready artifact for researchers, policymakers, and developers.

Empirically, the analysis confirms convergence toward risk-based device regulation and GDPR-inspired data protection, while highlighting persistent gaps in lifecycle surveillance, transparency mandates, and liability clarity—especially outside the most resourced agencies. The EU AI Act stands out as a binding inflection point for high-risk healthcare AI [14]. The United States illustrates high innovation throughput under a more sectoral governance mix [4]. Emerging economies often legislate privacy before AI-specific SaMD capacity fully materializes [24].

6.2 Limitations and Areas for Improvement

Limitations include subjectivity in expert scoring, incomplete cross-lingual coverage, heterogeneous device-count reporting, and the still-early stage of supervised NLP automation. Context constraints—academic hosting without an industry internship supervisor, public-data only, and zero cloud budget—bounded the depth of primary-law review in some jurisdictions and deferred live monitoring.

Areas for improvement follow directly: dual annotation for score reliability; inclusion of non-English primary sources; sub-national EU/US profiles; and formal usability testing with compliance officers. The dashboard remains a decision-support research tool, not a substitute for qualified legal counsel or notified-body advice.

6.3 Perspectives and Future Work

Concrete next steps include:

1. building a labeled clause corpus and evaluating transformer classifiers with reported F1 against HITL gold labels [30], [34];
2. adding longitudinal score versioning keyed to legislative events;
3. expanding country coverage and sub-national U.S./EU profiles;
4. integrating authenticated regulatory RSS/API feeds with change alerts;
5. conducting formal usability studies with regulators and medtech compliance officers;
6. exploring privacy-preserving analytics if de-identified hospital compliance logs become available under a DPIA;
7. packaging PDF/A archival exports and automated regression tests for table/figure regeneration.

These directions preserve the project's core ethic: automation may accelerate review, but human expertise remains responsible for compliance claims.

6.4 Answers to the Research Question

Returning to the research question in Chapter 1: multi-jurisdictional AI healthcare requirements *can* be systematically synthesized by combining thematic literature organization with a stable scoring ontology; they *can* be scored with transparent ordinal rubrics tied to primary instruments; and they *can* be interactively compared through open dashboards that preserve drill-down to narratives. The approach does not eliminate legal complexity, but it reduces the cost of first-pass comparative intelligence and creates a substrate for future NLP automation under HITL control.

6.5 Transferable Lessons

Three transferable lessons emerge for AI and data-science practice: (1) governance problems benefit from explicit ontologies as much as predictive problems benefit from feature schemas; (2) interactive analytics can be a first-class research contribution when paired with reproducible data; (3) responsible AI engineering includes knowing when automation must stop and expert review must begin.

6.6 Closing Statement

Governing AI in healthcare requires comparative intelligence that is rigorous enough for academic scrutiny and usable enough for professional briefing. By coupling a positioned literature synthesis with open, interactive analytics, this project provides a reproducible foundation for that intelligence—and a clear roadmap for turning curated knowledge into continuously maintained regulatory awareness.

Bibliography

[1] E. J. Topol, "High-performance medicine: The convergence of human and artificial intelligence," *Nature Medicine*, vol. 25, pp. 44–56, 2019.

[2] P. Rajpurkar, E. Chen, O. Banerjee, and E. J. Topol, "Ai in health and medicine," *Nature Medicine*, vol. 28, pp. 31–38, 2022.

[3] E. Wu, K. Wu, R. Daneshjou, et al., "How medical AI devices are evaluated: Limitations and recommendations from an analysis of FDA approvals," *Nature Medicine*, vol. 27, pp. 582–584, 2021.

[4] U.S. Food and Drug Administration, *Artificial intelligence/machine learning (ai/ml)-based software as a medical device (samd) action plan*, `https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-software-medical-device`, 2021.

[5] S. Gerke, T. Minssen, and I. G. Cohen, "Ethical and legal challenges of artificial intelligence-driven healthcare," *Artificial Intelligence in Healthcare*, pp. 295–336, 2020.

[6] D. S. Char, N. H. Shah, and D. Magnus, "Implementing machine learning in health care — addressing ethical challenges," *New England Journal of Medicine*, vol. 378, no. 11, pp. 981–983, 2018.

[7] M. Ghassemi, L. Oakden-Rayner, and A. L. Beam, "The false hope of current approaches to explainable AI in health care," *The Lancet Digital Health*, vol. 3, no. 11, e745–e750, 2021.

[8] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.

[9] I. G. Cohen and M. M. Mello, "HIPAA and protecting health information in the 21st century," *JAMA*, vol. 320, no. 3, pp. 231–232, 2018.

[10] B. Goodman and S. Flaxman, "European union regulations on algorithmic decision-making and a "right to explanation"," *AI Magazine*, vol. 38, no. 3, pp. 50–57, 2017.

[11] International Medical Device Regulators Forum, *Software as a medical device: Possible framework for risk categorization and corresponding considerations*, 2014.

[12] Pharmaceuticals and Medical Devices Agency (Japan), *Dash for samd: Regulatory framework*, 2020.

[13] National Medical Products Administration (China), *Guiding principles for registration review of AI medical device software*, 2021.

[14] European Parliament and Council, *Regulation (eu) 2024/1689 — artificial intelligence act*, 2024.

[15] U. J. Muehlematter, P. Daniore, and K. N. Vokinger, "Approval of artificial intelligence and machine learning-based medical devices in the USA and Europe (2015–20)," *The Lancet Digital Health*, vol. 3, no. 3, e195–e203, 2021.

[16] B. Babić, S. Gerke, T. Evgeniou, and I. G. Cohen, "Algorithms on regulatory lockdown in medicine," *Science*, vol. 366, no. 6470, pp. 1202–1204, 2019.

[17] T. J. Hwang, A. S. Kesselheim, and K. N. Vokinger, "Lifecycle regulation of AI- and ML-based software devices in medicine," *JAMA*, vol. 322, no. 23, pp. 2285–2286, 2019.

[18] U.S. Food and Drug Administration, *Marketing submission recommendations for a predetermined change control plan for ai/ml-enabled device software functions*, 2023.

[19] FDA, Health Canada, and MHRA, *Good machine learning practice for medical device development: Guiding principles*, 2021.

[20] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, "Dissecting racial bias in an algorithm used to manage the health of populations," *Science*, vol. 366, no. 6464, pp. 447–453, 2019.

[21] L. Seyyed-Kalantari, H. Zhang, M. B. A. McDermott, I. Y. Chen, and M. Ghassemi, "Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations," *Nature Medicine*, vol. 27, pp. 2176–2182, 2021.

[22] National Institute of Standards and Technology, *Artificial intelligence risk management framework (AI RMF 1.0)*, 2023.

[23] A. Jobin, M. Ienca, and E. Vayena, "The global landscape of AI ethics guidelines," *Nature Machine Intelligence*, vol. 1, no. 9, pp. 389–399, 2019.

[24] World Health Organization, *Ethics and governance of artificial intelligence for health*, Geneva, 2021.

[25] B. Mittelstadt, "Principles alone cannot guarantee ethical AI," *Nature Machine Intelligence*, vol. 1, no. 11, pp. 501–507, 2019.

[26] J. Morley, L. Floridi, L. Kinsey, and A. Elhalal, "From what to how: An initial review of publicly available AI ethics tools, methods and research," *Science and Engineering Ethics*, vol. 26, pp. 2141–2168, 2020.

[27] C. Petersen, J. Smith, R. R. Freimuth, et al., "Recommendations for the safe, effective use of adaptive CDS in the US healthcare system," *JAMIA*, vol. 29, no. 4, pp. 677–684, 2022.

[28] W. N. Price, S. Gerke, and I. G. Cohen, "Potential liability for physicians using artificial intelligence," *JAMA*, vol. 322, no. 18, pp. 1765–1766, 2019.

[29] OECD, *Oecd.ai policy observatory*, `https://oecd.ai/`.

[30] A. Vaswani et al., "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[31] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, 2015.

[32] A. Esteva et al., "A guide to deep learning in healthcare," *Nature Medicine*, vol. 25, pp. 24–29, 2019.

[33] G. Litjens et al., "A survey on deep learning in medical image analysis," in *Medical Image Analysis*, Journal survey commonly cited alongside conference imaging literature, vol. 42, 2017, pp. 60–88.

[34] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, 2019, pp. 4171–4186.

[35] I. Beltagy, K. Lo, and A. Cohan, "SciBERT: A pretrained language model for scientific text," in *Proc. of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019, pp. 3615–3620.

[36] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA: MIT Press, 2016.

[37] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson, 2020.

[38] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations (ICLR)*, 2021.

[39] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

A. Appendices

A.1 Appendix A — Significant Code Extracts

```json
{
  "country": "European Union",
  "region": "Europe",
  "maturity_level": "Advanced",
  "themes_scores": {
    "data_privacy": 10,
    "clinical_validation": 9,
    "approval_process": 9,
    "transparency": 9,
    "ethics": 10,
    "post_market": 9,
    "liability": 8
  }
}
```

Listing A.1: Illustrative JSON country record fields

```python
def composite_score(themes_scores: dict) -> float:
    vals = list(themes_scores.values())
    return sum(vals) / len(vals)
```

Listing A.2: Composite score computation (conceptual)

Full application code is maintained in the AI Clinic repository (`app.py`, `web/`, data generation scripts). Provide the Git URL in the final submission package.

A.2 Appendix B — Additional Data Notes

Theme definitions, source lists, and trend objects are stored alongside country records in `compliance_dataset.json`. Regenerated LaTeX tables used in Chapter 5 live under `generated/`.

A.3 Appendix C — Technical Environment

Recommended environment: Python 3.9+, Node.js 18+ for the React app, MiKTeX or TeX Live for report compilation. Core Python libraries include `streamlit`, `plotly`, `pandas`, and PDF export utilities. Hardware requirements are modest (CPU laptop sufficient for the dashboard MVP; GPU optional for future NLP fine-tuning).

A.4 Appendix D — Declaration of Generative AI Use

In accordance with aivancity PFE guidelines on generative AI:

- Generative AI assistants (including Cursor-based coding agents) were used to help structure LaTeX sources, expand draft wording from existing project materials, and accelerate formatting.
- All analyses, methodological choices, score interpretations, and conclusions were reviewed and validated by the student against primary regulatory sources and the curated dataset.
- AI-assisted code was inspected, tested locally where applicable, and remains the student's responsibility to understand and maintain.
- No AI system was used as a substitute for expert regulatory judgment; human-in-the-loop verification is part of the project design itself.